A cellular prior enables compact and transferable single-cell inference with CellPFN

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Abstract

Single-cell prediction often requires learning from limited labelled context across heterogeneous experiments. We developed CellPFN, a prior-data fitted network trained on a procedural cellular prior varying biological signals, donor structure and measurement noise. The 12.33-million-parameter model uses task-local gene correspondence without persistent gene identities or target-task gradient updates. After 384,000 scheduled synthetic episodes, CellPFN reached 72.9% mean balanced accuracy across four cell-typing panels and three support budgets, compared with 66.8–70.1% for released TabICLv2 and Kumo large under their tested input configurations. Cellular continued training improved three tabular models' one-shot balanced accuracy by 5.6–10.8 percentage points. Ablations revealed support-dependent benefits and tradeoffs. CellPFN also accepted transcriptome-scale inputs and yielded gene-deletion explanations recovering established cell-type markers. **The framework further identifies neuropathology-associated cellular states and reproducible predictive expression signatures across independent cohorts.**[O04] These results support cellular task priors for learning biological inference with limited labelled data.

Single-cell atlases provide increasingly broad reference maps, yet annotating a new experiment often requires inference from a small labelled set collected under different biological and technical conditions. More profiled cells do not necessarily supply more independent donors or more varied experimental settings. In supervised single-cell annotation, increasing donor diversity can contribute more to generalization than increasing cell count alone[1]. This distinction motivates models that learn how to interpret limited experimental context across heterogeneous tasks.

Single-cell foundation models such as Geneformer, scGPT, scFoundation, UCE and TranscriptFormer learn reusable representations from large collections of measured profiles[2–6]. Their breadth

supports transfer across biological settings, but large-scale pretraining does not guarantee an advantage on each downstream task. Evaluations have found limitations in frozen representations and saturation as pretraining collections expand[7,8]; recent preprint evidence also shows that normalization and linear methods can remain competitive across several benchmarks[9]. These findings motivate a complementary question: can learning improve the inference rule that combines a few labelled examples, even when the expression measurements themselves already provide useful features?

Meta-learning already supports transfer across heterogeneous single-cell experiments, as demonstrated by MARS[10]. Prior-data fitted networks (PFNs) provide a complementary way to learn inference rules. They are trained across prediction problems drawn from a generative prior, amortizing support-conditioned inference into a forward pass[11]. TabPFN and TabICL demonstrate that synthetic task distributions can produce effective general-purpose tabular learners[12–15]. Recent MapPFN and PerturbPFN work further applies prior-based learning to cellular perturbation prediction[16,17]. For transcriptomic classification, the opportunity is to encode the variation encountered across cellular experiments directly into the training prior, while accommodating thousands of sparse features and scarce labels.

Single-cell simulators already model count distributions, dependence, donor variation and regulatory dynamics through complementary statistical and mechanistic approaches[18–25]. Learned generative models also support conditional synthesis and augmentation[26,27]. For PFN training, the design objective extends beyond reproducing a reference dataset to generating diverse labelled inference problems. Controlled variation in signal strength, hidden subpopulations, donor effects and measurement noise can expose a learner to experimental configurations without requiring each configuration to be measured and annotated. Synthetic-to-real learning in optical flow and face analysis provides precedents for learning from known simulated structure[28,29]. Nevertheless, simulator fidelity depends on the statistic and scenario being evaluated[30,31]. The relevant test is whether the induced prior teaches inference that transfers to real biological data.

Here we introduce CellPFN and its procedural cellular prior. CellPFN preserves gene correspondence within each task without persistent gene identities across tasks. We test whether a compact model can achieve competitive biological inference with limited synthetic training exposure, and whether the same simulator can improve released tabular models through continued training. Quantitative ablations examine which architectural and task-distribution choices help, where they fail, and how simulator statistics relate to transfer. We further evaluate transcriptome-scale inputs and query-gene explanations, grounding the prior's utility in inference from limited labelled biological context.

Results

A cellular prior teaches inference from labelled context

We constructed a procedural prior over single-cell prediction problems, varying biological signals together with their observation conditions. The simulator delivers PyTorch-compatible cell streams

and reusable biological datasets to both learning routes (Fig. 1). The generator samples hierarchical identities, distributed expression programs, rare markers, correlated variation and donor-dependent states. Donor composition, study and batch shifts, sequencing depth and overdispersed count sampling produce heterogeneous observations. Varying labels and support budgets turns each simulated dataset into multiple inference problems with known labels and controllable nuisance factors. Aggregate statistics from real datasets, including those evaluated here, informed the prior's ranges. All examples used to optimize CellPFN's weights were synthetic.

Each problem comprises labelled support cells and unlabelled queries. Genes retain their correspondence within a task but have no persistent identity across tasks. CellPFN compares each query gene with support-derived class summaries, pools these relations through permutation-invariant statistics[32] and integrates them into direct neural class predictions. A covariance estimate fitted to the support supplies an additional statistical channel when support is sufficient. Prediction requires no gradient updates, and queries do not condition on other query cells.

The reference model has 12.33 million trainable parameters and received 384,000 scheduled episodes from 12,000 simulated dataset banks; 276,828 episodes carried nonzero loss weight. Reused banks supplied distinct task views. TabICLv2 reports approximately 35 million synthetic pretraining datasets[15], about 90 times CellPFN's scheduled episode count (Supplementary Table S2). These different exposure units contextualize training scale but do not establish a matched-compute advantage.

Compact CellPFN is competitive across biological inference regimes

Across two biological collections, we evaluated eight-class coarse and twenty-class fine peripheral blood mononuclear cell (PBMC) annotation[33], and four-class ZEBRA cortex annotation in development and held-study panels[34]. The cortex classes were glia, immune, neural and vascular cells. Each panel used one, ten or 100 labelled support cells per class with three matched support draws. Comparators included statistical classifiers, released tabular models, and logistic classifiers on Geneformer-V2 and scGPT embeddings[2,3,15,35–37]. Each model retained its declared input route (Methods; Fig. 2a–d).

Across the four panels and three support sizes, all-gene CellPFN reached 72.9% mean balanced accuracy, compared with 66.8% for released TabICLv2 and 70.1% for Kumo large using 2k highly variable genes (HVGs; Table 1). Geneformer reached 74.9% through its native input route. These averages indicate competitive performance, without establishing equivalence or universal superiority. Supplementary Fig. S2 and Table S1 reports panel-specific means and three-draw ranges.

Conventional low-rank linear discriminant analysis (LDA) remained strong with abundant labels: at 100 supports per class, its four-panel mean was 84.6%, compared with 85.4% for CellPFN; at ten supports, the means were 75.3% and 79.6%. LDA used processed expression and was not evaluated at one support.

Performance varied substantially across tasks. For fine PBMC annotation at one support, Geneformer reached 68.8% versus CellPFN's 51.1%; for held-out cortex at ten supports, CellPFN reached 91.9% versus Geneformer's 87.4%. Both foundation models likely encountered the PBMC

cells during unlabelled pretraining. Geneformer-V2 corpus records overlap studies supplying 92.6% of held-out cortex queries, an upper bound on exact-cell overlap. This evidence does not establish label exposure or explain the ranking (Table 1). Model variants were selected on these development panels, with one training seed per arm. **A frozen evaluation on additional, independently held-out cohorts will determine how consistently this competitive performance generalizes.**[O01]

Finer PBMC distinctions remained difficult. With up to 8,192 HVGs, a separate evaluation gave one-shot balanced accuracies of 62.3% for eight coarse types, 53.9% for 20 fine types and 32.4% across eight harder tasks. The latter separated sibling CD4, CD8, B, NK or broader T-cell subtypes, or all 35 fine-grained labels; their means rose to 63.9% and 80.2% at ten and 100 supports (Supplementary Table S9). Nine CD4 subtypes yielded 23.4% and 79.6% at one and 100 supports. This analysis used ten support draws at one/ten supports and five at 100. These post hoc tasks are not a prespecified difficulty stratum; their means lack draw-level intervals. Two finer cortical tasks yielded one-shot accuracies of 67.8% for seven development labels and 52.5% for three held-study neural labels (Supplementary Table S9). CellPFN handled twenty and 35 PBMC labels despite training with at most eight classes.

The cellular prior also improves pretrained tabular models

The cellular prior also improved pretrained tabular learners (Fig. 2e–g). We continued training released TabICLv2, Kumo small and Kumo large using the cellular prior throughout, with 40,000 cellular tasks and 10,000 generic replay tasks over 2,500 updates per model. At 2k HVGs, one-shot balanced accuracy improved by 7.3, 10.8 and 5.6 percentage points, respectively, averaged over four panels (Table 1). Changes at 100 supports were below one point in magnitude. Benefits were not uniform: fine PBMC accuracy for TabICLv2 at ten supports fell from 73.8% to 64.6%, with all three support draws worsening (Supplementary Table S2).

Cellular continued training of Kumo small improved accuracy by 6.1 points across eight low-support panel–budget combinations, compared with approximately zero change after generic-table continued training. Two further cellular task streams yielded gains of 5.7 and 5.2 points with the same model seed. The control differed in rows and columns per update, so task content was not isolated under matched data volume. Adaptation could also reduce generic-table performance: TabICLv2's one-shot mean balanced accuracy across four generic tables and five split seeds fell from 70.9% to 64.0% (Supplementary Table S2).

Gene-anonymous inference extends to transcriptome-scale inputs

CellPFN processed 33,538 PBMC and 21,158 cortex genes after training with at most 8,192. Shared per-gene computations and symmetric aggregation required no expanded vocabulary. Across the three benchmark draws, balanced accuracy was 71.2% with 2k HVGs, 73.8% with 8,192 and 72.9% with all genes (Table 2). A separate twenty-draw analysis resolved task-specific effects: all genes versus 2k HVGs improved held-out cortex accuracy at ten supports by 7.4 points (95% interval, 5.3–9.6), but reduced one-shot PBMC accuracy by 3.3 points for coarse and 2.0 points for fine labels

(Supplementary Fig. S3 and Table S3). These paired intervals quantify support-draw variation, not training-run variation.

Released TabICLv2 declined from 66.8% to 48.3% and Kumo large from 70.1% to 66.4% when moving from 2k HVGs to all genes. TabPFN-3.5 refused the all-gene cases at its default feature guard[37,38]. These are outcomes of the tested column-handling configurations, not fundamental architectural limits. Random-gene controls and alternative input views are reported in Supplementary Tables S3 and S9.

Prior and architecture changes have regime-dependent effects

The support-fitted covariance channel had substantial but regime-dependent effects. On fine PBMC annotation at 100 supports, separately trained models with no operator, diagonal variance information or the full operator reached 82.2%, 85.7% and 89.0% balanced accuracy. Full versus no operator gained 6.81 points (95% paired interval, 6.36–7.26); full versus diagonal gained 3.37 points (3.06–3.68), over twenty support draws with one training run per arm (Supplementary Fig. S4 and Table S4). The prespecified operator test did not meet its retention criterion: the full model lost 1.56 points against the no-operator model on coarse PBMC at ten supports, exceeding the allowed 1.50-point harm threshold. The descriptive gains do not overturn that outcome.

A ten-candidate prior screen identified task-mixture restriction as the only change meeting its retention rule on the Kumo-small proxy (Supplementary Fig. S5 and Table S5). After 10,000 cellular tasks, low-support accuracy improved by 1.95 points, with gains of 1.63, 1.62 and 2.61 across three task streams sharing one model seed. No prior-content modification demonstrated benefit. Whole-study gene absence, for example, gained 0.90 points but failed its predictive-log-loss guard, worsening negative log likelihood (NLL) by 0.099 against a 0.03 limit. Other outcomes included neutral and unresolved cases; they do not establish that the corresponding mechanisms are irrelevant.

The restricted mixture also improved CellPFN's one-shot accuracy on all four panels: gains were 1.11, 2.42, 2.56 and 3.49 points for coarse PBMC, fine PBMC, development cortex and held-out cortex, respectively (Supplementary Fig. S5 and Table S6). The held-out cortex gain had a 95% paired interval of 1.73–5.26 over twenty support draws. At 100 supports, NLL worsened on three panels by 0.35–0.63, while improving by 0.05 on fine PBMC. The restriction replaced an approximately 80% metadata-task share with 75% cell-type and 25% donor-identity tasks and excluded signed-covariance banks. It also processed 1.67 times as many cells per update. The accuracy and log-loss changes therefore reflect a combined task-mixture intervention.

Simulator diagnostics likewise supported partial agreement with real data (Supplementary Fig. S1 and Table S7). The median strongest-marker absolute log2 fold change was 6.23 under the cellular prior versus 7.10 in the real reference and 1.94 under the baseline prior. Yet eight-class tasks with ten supports were easier: median chance-adjusted nearest-class-mean accuracy was 0.881 versus 0.775, and 37.5% of synthetic banks exceeded the real maximum. These reference statistics informed prior calibration, and the cellular prior supplied only the final 64,000 CellPFN episodes. They therefore

contextualize transfer without establishing that increased realism caused better predictions.

Sex prediction varies across evaluation settings

CellPFN also predicted sex in ZEBRA, but performance depended on the evaluation setting. On one frozen oligodendrocyte case, it achieved 100% balanced accuracy over 50 query cells and over ten query GSM samples, using 100 support cells per sex. Support and query GSMs were disjoint; GSM identifiers do not guarantee distinct donors. A later positive control restricted studies to those measuring *XIST* and at least four Y-linked genes. With up to 8,192 HVGs, mean unit-level AUC was 0.977 at four support units per sex and 1.000 at 16, across ten splits; units included age-by-sex sample groups. The broader sex task yielded 0.507 and 0.538 at those budgets (Supplementary Table S9). Study sets and gene universes also differed, so this contrast does not isolate the effect of sex-linked genes. **Gene-level attribution of these sex predictions will establish which expression features the model uses; the available results do not yet identify those features.**[O06]

Query-gene deletion recovers cell-type marker structure

Gene anonymity permits gene-level explanations because names can be restored after prediction (Fig. 3). DeepSets leave-one-gene-out (DS-LOO) approximates the effect of zeroing one query gene on the logit margin between the predicted class and its original runner-up, holding support and fitted covariance fixed. Unlike gradient attribution along an input path[39], the target is a finite deletion. One attribution sweep combines pooled gene statistics and operator scalars with head-gradient evaluations and depth/detection corrections, avoiding a separate full-model forward pass per deleted gene. It is not literally one forward pass, and its target is a model intervention rather than a biological knockout.

Against exact deletions in six cases and two model states, case-level medians of within-cell rank correlation ranged from 0.979 to 0.998; reference-model relative L2 errors were 0.5–4.9%. Validation used selected genes and correctly classified cells in full precision (Supplementary Fig. S8 and Table S8). **All-gene explanations cost approximately 10–13 single-gene exact forward passes. For the PBMC example, DS-LOO took 67.9 seconds; the corresponding 61-hour exhaustive-deletion runtime was extrapolated from individual forward timings. Shared-GPU timings remain provisional.**[O05]

The explanations recovered recognizable cell-type markers without gene-name embeddings. In PBMC class-margin rankings at 100 supports, B-cell markers *CD79A* and *MS4A1* ranked first and second; *IL7R* ranked second for CD4 T cells, *CD8A* second for CD8 T cells, and *GNLY*/*NKG7* second/seventh for natural killer cells[33,40]. In development cortex, oligodendrocyte-associated *PLP1* ranked eighth for glia, neuronal *SYT1* seventh for neural cells, and endothelial *CLDN5* sixteenth for vascular cells[41]. These are selected lineage-associated matches, not an enrichment test (Supplementary Table S8).

Removing each PBMC query's 50 highest-ranked genes reversed its original runner-up margin in

45.0% of cells, versus 0.3% for random expressed genes. Class-level top-20 overlap across support configurations was 0.43–0.56, indicating context sensitivity. Thus the rankings explain classifier decisions without establishing perturbation responses or universal marker recovery.

A provisional application examines neuropathology-associated expression

In a provisional donor-level application to ROSMAP-derived single-nucleus data[42], microglial predictions distinguished CERAD 4-versus-1 neuropathology groups with an AUC of 0.745 and a reported corrected 95% repeated-split interval of 0.700–0.790. Case and control support was matched within sex (Supplementary Table S9). The result awaits a donor-label audit and frozen rerun; it measures association, not prospective diagnosis or clinical utility.[O02]

Applying query-gene deletion to the corrected disease analysis and testing the findings in independent cohorts will test whether this approach can connect donor-level prediction to reproducible cellular expression signatures.[O03]

Table 1: Biological classification and cellular-prior transfer.

Model comparison: balanced accuracy (%)

Panel, support/class	CellPFN	TabICLv2		Kumo small		Kumo large		TabPFN 3.5	Geneformer V2 (316M)	scGPT	Low-rank LDA
	All genes	Released	+ prior	Released	+ prior	Released	+ prior	Released	+ LR	+ LR	All genes
PBMC coarse, 1	60.1	50.2	62.3	45.8	**62.8**	58.9	62.2	34.3	**67.1***	64.9*	–
PBMC coarse, 10	84.0	81.4	**85.2**	82.2	83.3	81.7	82.5	82.6	**88.5***	77.5*	79.4
PBMC coarse, 100	91.9	92.4	**93.4**	92.6	92.3	92.8	92.8	91.5	92.7*	80.7*	90.0
PBMC fine, 1	**51.1**	30.0	29.1	44.1	49.0	46.3	50.9	33.0	**68.8***	55.7*	–
PBMC fine, 10	79.0	73.8	64.6	80.0	80.5	**83.3**	82.2	79.3	**84.9***	65.9*	70.3
PBMC fine, 100	89.0	83.6	85.3	88.3	88.7	**89.3**	89.2	86.9	**90.4***	72.0*	87.8
ZEBRA development, 1	41.4	39.5	44.6	31.2	41.9	40.1	42.8	38.4	43.1$^{(*)}$	**47.6**	–
ZEBRA development, 10	**63.6**	59.9	60.6	57.5	59.2	56.8	60.6	59.3	61.1$^{(*)}$	60.2	60.6
ZEBRA development, 100	**66.1**	65.3	65.3	65.1	64.9	65.1	65.8	65.5	65.4$^{(*)}$	65.8	66.0
ZEBRA held-study, 1	62.8	49.8	62.7	48.8	59.3	49.1	60.8	54.1	58.4*	**65.5**	–
ZEBRA held-study, 10	**91.9**	83.2	85.8	84.8	86.8	85.1	85.9	85.3	87.4*	87.2	91.0
ZEBRA held-study, 100	**94.7**	92.9	93.2	93.7	92.9	93.1	93.4	93.9	90.3*	89.5	**94.7**
Overall	**72.9**	66.8	69.4	67.8	71.8	70.1	72.4	67.0	**74.9***	69.4*	–

Means over three support draws. Bold identifies the highest row mean overall and among unflagged entries, including ties at displayed precision; gray marks starred exposure flags. Rankings are descriptive, not significance claims. Overall equally weights twelve panel–budget combinations; dashes denote unevaluated settings, and LDA has no overall mean. PBMC coarse and fine panels have eight and twenty classes; both ZEBRA panels classify four broad groups (Glia, Immune cell, Neural and Vascular cell), with distinct development and held-study query panels. Model variants were selected using development evaluations. Tabular models use 2k HVGs, a requested cap of up to 2,000 support-selected genes; actual widths can be lower (Methods). CellPFN and LDA use all genes. CellPFN uses PBMC counts and processed ZEBRA values; tabular models and LDA use processed values. Geneformer/scGPT use native embeddings with logistic regression (LR); scGPT uses the declared unstandardized route, with post hoc classifier sensitivity in Table S9. "+ prior" denotes released weights adapted with 40,000 cellular tasks plus generic replay. Panel-specific draw ranges appear in Table S1. HVG, highly variable gene.

* flags likely query-source pretraining overlap: PBMC query cells are assessed as likely present for Geneformer-V2 (316M) [35] and scGPT. Geneformer-V2 corpus records overlap source studies supplying 92.6% of ZEBRA held-study query cells; this is an upper bound on exact-cell overlap, not verified cell matching. $^{(*)}$ denotes inferred possible overlap of control query cells in ZEBRA development (at most 22.3%). Support-source overlap is audited separately and is not encoded by these query-source stars. These marks concern cell-expression exposure, not evidence that target labels were seen. Overall starred means include flagged evaluations. Unmarked scores do not certify absence of exposure within or outside the audited Geneformer/scGPT corpora.

Table 2: CellPFN gene-panel ablation.

Balanced accuracy (%)

Panel	1 support/class			10 supports/class			100 supports/class		
	2k HVGs	8,192 HVGs	All genes	2k HVGs	8,192 HVGs	All genes	2k HVGs	8,192 HVGs	All genes
PBMC coarse	60.4	**61.1**	60.1	82.6	82.8	**84.0**	90.6	**92.0**	91.9
PBMC fine	53.0	**54.3**	51.1	79.8	**81.3**	79.0	87.4	**90.3**	89.0
ZEBRA development	43.2	**45.8**	41.4	60.8	63.4	**63.6**	64.2	65.1	**66.1**
ZEBRA held-study	61.1	**65.0**	62.8	80.9	**92.0**	91.9	89.8	92.7	**94.7**

Means over three support draws for the same CellPFN checkpoint and query panels used in Table 1. Bold identifies the highest mean within each panel and support-budget group, including ties at displayed precision. Rankings are descriptive, not significance claims. The labels 2k HVGs and 8,192 HVGs denote requested caps of up to 2,000 and 8,192 support-selected genes; actual widths can be lower (Methods). CellPFN uses PBMC counts and processed ZEBRA values; ZEBRA development and held-study query panels are distinct. HVG, highly variable gene. Table S3 reports the separate twenty-draw analysis, paired intervals and random-gene controls.

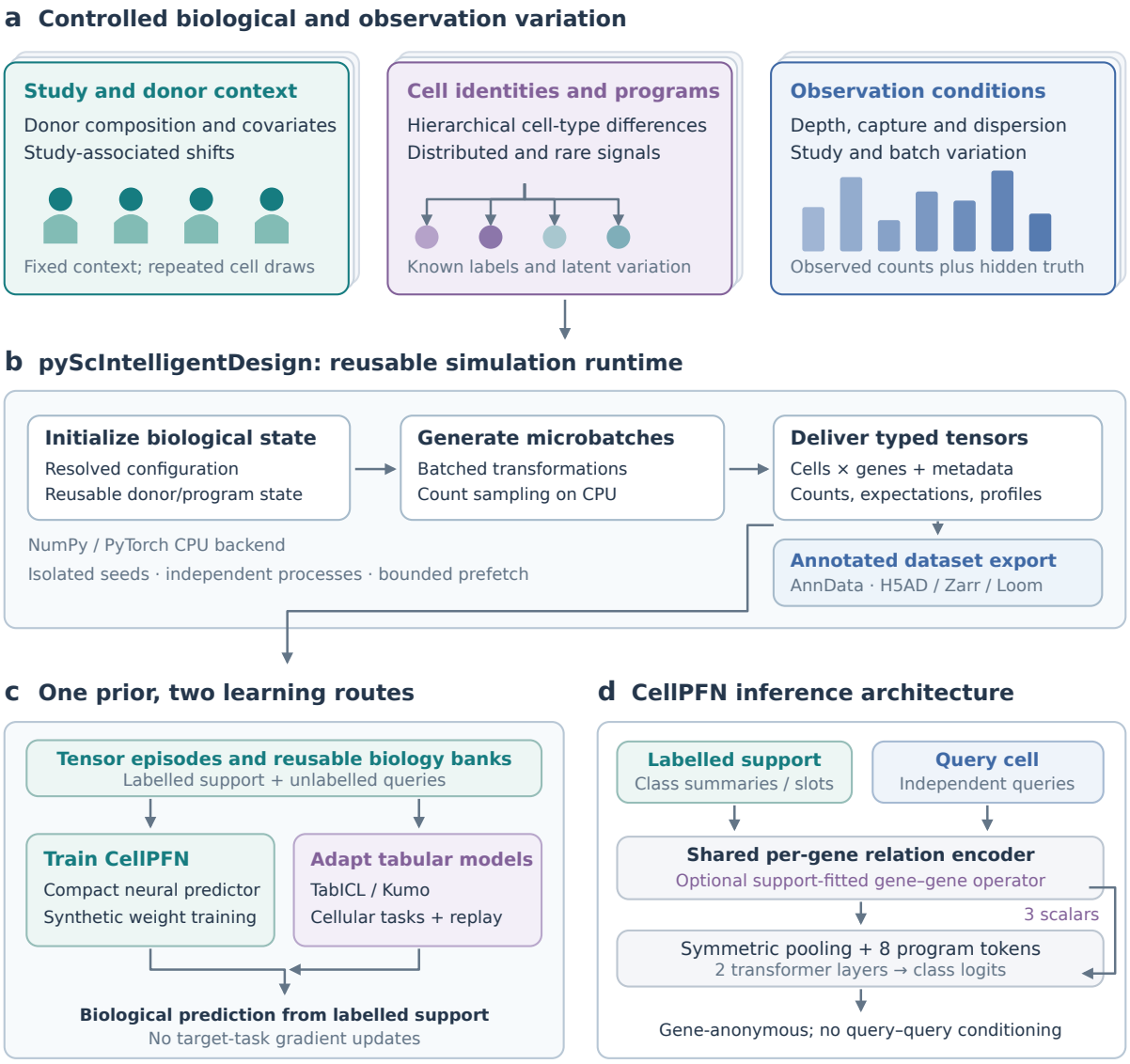
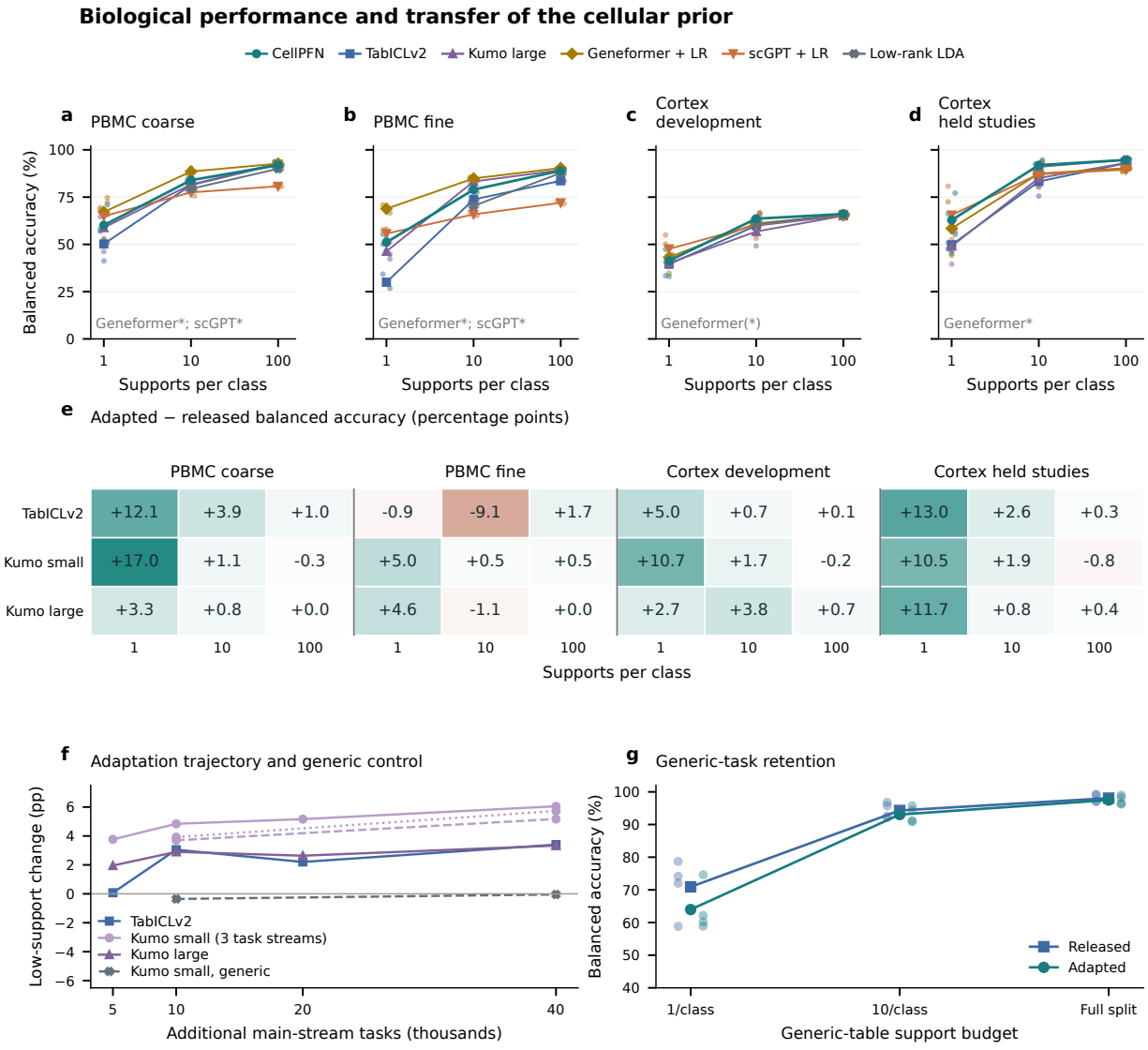


Figure 1: A reusable cellular simulator supplies two learning routes. a, Configurable donor context, cell identities and expression programs are combined with observation conditions to generate cells with known labels and latent truth. **b,** pyScIntelligentDesign separates initialization of biological state from repeated microbatch generation. CPU computation combines NumPy and PyTorch; typed tensors connect generation to training, while annotated exports support independent dataset use. Independent producer processes and bounded prefetch overlap task generation with training. Hidden profiles and expectations support diagnostics and are excluded from classifier inputs. **c,** Tensor episodes and reusable biology banks train compact CellPFN or continue training released tabular models with cellular tasks and generic replay. Both applications perform downstream inference from labelled support without target-task gradient updates. **d,** CellPFN compares each query with support-derived gene summaries and learned slots, optionally incorporating a support-fitted gene–gene operator through a gene channel and three scalars supplied to the prediction head. Symmetric aggregation and learned program tokens feed a transformer and class prediction head. The tokens are computational summaries, not annotated biological pathways. The schematic describes architecture and data flow; shapes do not represent measured quantities.



CellPFN: 12.33M parameters · 384k scheduled episodes · 276,828 loss-bearing episodes
Adaptation: 40k cellular + 10k replay tasks; released pretraining is additional. * / (*) mark query-source exposure evidence.

Figure 2: Biological performance and transfer of the cellular prior. a–d, Balanced accuracy for coarse/fine PBMC and development/held-study cortex. Small points show three support draws; connected symbols show means for fixed queries and one training seed. CellPFN/LDA use all genes, tabular models 2k HVGs, and Geneformer/scGPT native embeddings plus logistic regression. One-shot LDA is unavailable. Gray panel annotations flag likely query-source exposure with stars; parenthesized stars indicate possible overlap (Table 1), without verified target-label exposure. **e,** Mean adapted-minus-released changes over three paired support draws, retaining negative changes. Released weights were rescored within each adaptation run; deltas need not equal differences between stored benchmark means. **f,** Mean changes across eight low-support panel–budget conditions. Kumo-small cellular trajectories vary task streams with one model seed. The generic-only control has unmatched row/column exposure. Cellular counts exclude 25% additional generic replay; released-model pretraining precedes adaptation. Task counts do not establish matched compute. **g,** Generic-task retention. Large symbols equally average four datasets and five split seeds; small dots show dataset means. "Full split" uses the available training partition. No uncertainty across tasks or queries is interpreted as training uncertainty. Benchmark panels informed development; they are not independent confirmatory cohorts. HVG labels denote requested caps (Methods).

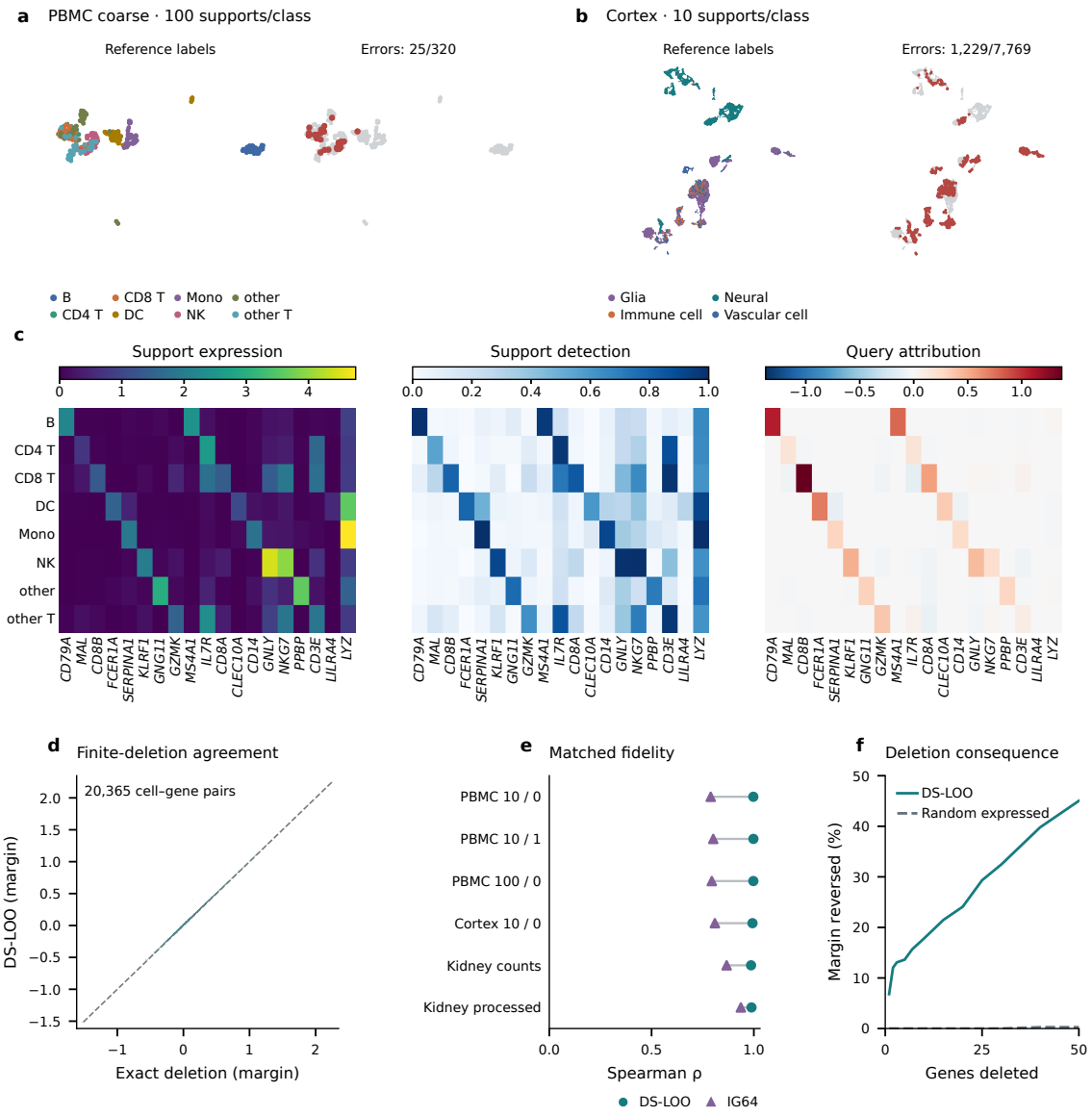


Figure 3: Gene-deletion explanations in biological context. a,b, RNA UMAPs show reference labels and prediction errors (red) for all 320 PBMC queries (100 supports/class) and 7,769 development-cortex queries (ten supports/class), both draw 0. Paired views share coordinates fitted without labels to the support/query union (Methods). **c,** PBMC support mean log1p(counts per 10,000), detection fraction and mean corrected DS-LOO query attribution. Independent scales distinguish expression, detection and classifier-margin units. The 19 genes combine each class's first-ranked gene and all marker checks in Supplementary Table S8. Positive attribution supports the predicted class relative to its fixed runner-up; negative attribution opposes that margin. **d,** All 20,365 exported PBMC cell–gene validation pairs, including 1,625 additional DS-LOO-selected pairs and retaining zero and negative effects; dashed line, identity. Exact deletion covers selected genes, not every gene. **e,** DS-LOO and IG64 fidelity on the same original evaluated gene set: within-cell Spearman correlations summarized within and across classes. Labels give supports/draw. Six configurations, including two kidney representations, are not six independent cohorts. **f,** Ranked deletion versus two pooled random-expressed orderings/cell reverses 45.0% versus 0.3% of original margins at 50 genes. Explanations use 191/320 PBMC and 96/7,769 cortex queries correctly classified in both stored and float32 predictions. Summaries are descriptive, without confidence intervals. Zeroing an absent gene cannot assess absence information; model-deletion fidelity does not establish biological causality.

Methods

CellPFN prediction task and expression representations

An episode consists of a labelled support matrix $X_S \in \mathbb{R}_{\geq 0}^{N \times G}$, support labels $y_S \in \{1, \ldots, C\}^N$ and a query matrix $X_Q \in \mathbb{R}_{\geq 0}^{Q \times G}$. The same G gene coordinates occur in support and query, but the network receives neither persistent gene identifiers nor donor, study or simulator metadata. Class names are localized to the support classes. CellPFN returns one logit for each query–class pair and class probabilities by softmax. Its parameters remain fixed during prediction; support fitting constructs episode-specific statistics and learned support summaries. Query cells share this support state but do not attend to one another. Consequently, changing the other query rows does not change a given row's prediction, apart from numerical effects of batching. The architecture is summarized in Fig. 1 and Supplementary Table S12.

Count and processed-expression inputs use separate linear input projections and a shared nonlinear backbone. For count inputs, the delivered-panel depth is $d_i = \max(1, \sum_g x_{ig})$, the composition is $p_{ig} = x_{ig}/d_i$, and the expression view is $\ell_{ig} = \log(1 + 10^4 p_{ig})$. Processed inputs must already contain nonnegative log-normalized expression; the supplied values are used as ℓ_{ig} without another normalization or logarithm. For either mode, we compute support-wide gene means $\bar{\ell}_g$, population standard deviations s_g and detection fractions, together with the corresponding statistics within each support class. We use $\epsilon = 10^{-6}$ and define the support-standardized view

$$z_{ig} = \mathrm{clip}\left(\frac{\ell_{ig} - \bar{\ell}_g}{\max(s_g, \epsilon)}, -8, 8\right). \tag{1}$$

The support statistics also transform query rows. This standardized log-expression view is an input feature, not a negative-binomial Pearson residual. Training assigns count and processed modes in a 70:30 mixture; processed training inputs are obtained by row-wise log-normalization over the complete delivered gene panel.

Cellular task prior and simulator architecture

pyScIntelligentDesign separates the definition of a synthetic experiment from the production of individual cells. A configuration specifies a distribution over gene abundance, cell populations, donor attributes and observation processes. Instantiating this distribution fixes the biological parameters of one experiment; repeated sampling then produces cells sharing its population and donor structure. The resulting prior supplies both CellPFN training episodes and the cellular tasks used to adapt tabular models (Fig. 1). The implementation also exposes an independent simulation interface for annotated single-cell datasets. We describe the cellular prior used in the principal training endpoint below; the baseline prior is a comparison specification, and optional software capabilities are distinguished in Supplementary Table S11.

Let G denote the number of anonymous genes, c_i the cell type, d_i the donor, and s_i and b_i the study and batch of cell i. An experiment contains a positive baseline composition $\boldsymbol{\pi}$ with

$\sum_g \pi_g = 1$, cell-type programs, donor metadata and shared nuisance parameters. Gene identities are procedural indices. A single permutation is applied consistently to all cells in a delivered task; no correspondence to named real genes is required. The experiment hierarchy has four studies with one batch per study and 16 generated donors. Fixed training banks use eight donors, eight cell types and 8,192 genes; bank construction and the subsequent selection of task targets are separate operations.

Biological variation and gene programs

The abundance distribution combines a lognormal body with a sparse high-abundance component, following the use of heterogeneous gene means and outliers in single-cell simulation [18]. It is normalized to a composition and constrained so that no gene exceeds 10% of baseline mass. Three biological task families generate distinct relationships between expression and cell type: distributed mean programs, intermittently expressed markers superimposed on stable mean programs, and signed covariance programs. The bank schedule allocates these families in a 2:1:1 ratio. Biological parameters and technical randomness have separate seed namespaces, permitting controlled changes to observation conditions without redefining the underlying experiment.

For mean-program tasks, sparse gene loadings are combined into class-dependent shifts, with a hierarchy producing shared variation among related classes. For a normalized gene-weight vector $\boldsymbol{w}$, the implementation centres a shift $\boldsymbol{v}$ by $v_g - \sum_h w_h v_h$ and scales its weighted squared magnitude to the configured energy. Class shifts use $w_g \propto \pi_g^{1/2}$; their final values are smoothly bounded by $L \tanh(v_g/L)$, with $L = 4.5$. A class template is proportional to $\pi_g \exp(v_{cg})$. A construction is admitted only if every class template has maximum gene share at most 0.35, with at most 16 construction attempts. These bounds apply to the class template, rather than imposing the same bound on every fully perturbed cell.

The marker family adds a class-specific sparse marker set. A Bernoulli draw controls whether the marker program is expressed in a cell, and a lognormal multiplier varies its amplitude. Marker and class shifts are combined before saturation, preserving the same class-shift ceiling. Hidden subtypes add within-class modes whose offsets are centred using their Dirichlet mixture weights. This permits multimodality without introducing an arbitrary unweighted class-mean shift. Optional trajectories vary position along class-centred program directions, with cell-specific position jitter; class-by-study and class-by-donor interactions provide domain-dependent changes in expression. Both trajectory and interaction blocks are independently enabled with probability 0.25 in the principal specification.

The covariance family isolates information carried by relationships between genes. Six sparse programs V_{jg} are arranged in three pairs. For each pair, u_j is a Rademacher variable and $e_j \in \{-1, 1\}$ has expectation ρ; the private class code supplies a sign $a_{cj} \in \{-1, 1\}$. The latent coordinates are $z_{2j-1} = u_j$ and $z_{2j} = a_{cj} u_j e_j$, so their marginal means are zero and their paired correlations have

class-dependent signs. The biological profile is

$$p_{ig}^{\text{bio}} \propto \pi_g \left(1 + \sum_{j=1}^{6} z_{ij} V_{jg}\right). \tag{2}$$

Program construction enforces positivity and weighted centring. Thus the class signal is placed in covariance at this biological layer; downstream nuisance transformations and count sampling remain part of the complete observation model.

Within-type cell state combines an independent residual with shared sparse programs and a weaker quality-associated direction. If r_{ig} is the energy-normalized residual and h_{ig} the centred structured state, the combined shift is

$$u_{ig} = \sqrt{1-f}\, r_{ig} + \sqrt{f}\, \kappa h_{ig} - o_g. \tag{3}$$

Here f is the structured fraction, κ is fixed once per experiment using a separate reference sample of 256 cells per class, and o_g partially compensates the mean introduced by exponentiating structured variation. Program genes are sampled with probabilities proportional to baseline abundance raised to an experiment-specific power. Programs can be shared across classes or restricted to one class. Their activation probabilities vary by donor through a Gaussian donor-by-program logit shift. Fixing the normalization reference before cell production prevents the state scale from depending on the requested batch size or cell order.

Donor metadata are generated using the simulator's vine-copula machinery. Continuous latent metadata are transformed to their public measurement scales, including age and sex. Centred, standardized metadata act through sparse anonymous gene directions, normalized by baseline-weighted energy; configurable nonlinear kernels introduce additional shared gene responses. Selected-target and other-metadata contributions are retained separately and multiplied without renormalizing the combined effect. These metadata generate labels and expression jointly; task labels are not imposed independently on a completed expression matrix. The distinction between donor-disjoint phenotype tasks and closed-set donor-identity tasks is maintained during episode construction.

The donor aging component adapts stochastic perturbations in normalized log-count coordinates [43]. A profile is placed at fixed reference depth 1,500, transformed by $q_g = \log(1+x_g)/\max_h \log(1+x_h)$, perturbed by an initial Gaussian increment and a sequence of clipped Gaussian increments, and transformed back. Chronological age maps from the fixed 5–110-year coordinate system to 0–105 steps; generated donor ages span 60–90 years. Initial and per-step standard deviations are 0.01 and 0.005, respectively, with a lognormal donor rate of mean one and log-standard deviation 0.2. Gene-specific rate mixtures and donor jitter vary susceptibility. A donor's entire clipped walk is prepared once as an equivalent offset and lower/upper clipping map, then reused for all its cells; this preserves the sequential clipping operation while amortizing the step loop. Before compositional renormalization, expression is bounded to e^{-1}–e^{1} times the unaged input. The direct metadata age direction is disabled when the clock is active. Separate configured components add directional age

shifts and age-dependent cell-noise scaling; the latter scales deviations around the expected log-state shift, preserving that log-state expectation rather than mean expression after exponentiation. Parameter distributions are given in Supplementary Table S10.

Observation model and calibration

The biological profile, metadata/aging effects, cell state, donor effects and study/batch effects are combined in log-expression space and normalized:

$$p_{ig} = \frac{p_{ig}^{\text{bio}} \exp(\Delta_{ig})}{\sum_h p_{ih}^{\text{bio}} \exp(\Delta_{ih})}. \tag{4}$$

Here Δ_{ig} comprises the enabled shifts, with the aging component inserted through the log-ratio of aged to unaged biology so that metadata are not applied twice. Library size has a log-additive study–batch–donor–cell hierarchy,

$$\log L_i = \ell_0 + \ell_{s_i}^{\text{study}} + \ell_{b_i}^{\text{batch}} + \ell_{d_i}^{\text{donor}} + \epsilon_i + \delta_0, \tag{5}$$

with experiment-specific scale multipliers and a lower library bound of 50. The normalized profile is converted to counts through a gamma–Poisson model:

$$A_{ig} \sim \text{Gamma}(\tilde{k}_{ig}, 1/\tilde{k}_{ig}), \qquad \lambda_{ig} = L_i p_{ig} A_{ig}, \qquad X_{ig} \mid \lambda_{ig} \sim \text{Poisson}(\lambda_{ig}). \tag{6}$$

The second gamma argument is scale, giving unit mean and variance $1/\tilde{k}_{ig}$. Conditional on L_i, p_{ig} and $\tilde{k}_{ig}$, integrating the gamma draw yields mean $L_i p_{ig}$ and variance $L_i p_{ig} + (L_i p_{ig})^2/\tilde{k}_{ig}$. The terminal Poisson sampler is applied exactly once.

Dispersion follows the realized class template where a class-mean template exists, and otherwise the baseline profile. At nominal mean m_{cg},

$$k_{cg} = \text{clip}\left[\frac{J_g}{\phi_0 + a \max(m_{cg}, m_0)^{-\alpha}},\ 0.01,\ 160\right], \tag{7}$$

where J_g is lognormal gene jitter, $\phi_0 = 0.1$, $m_0 = 0.1$, $\alpha = 1.2$, and a is drawn log-uniformly from 0.05–2.0. For the enabled age-dependent noise multiplier m_{ig}^{age}, the sampler uses the effective shape $\tilde{k}_{ig} = \text{clip}[k_{c_i g}/(m_{ig}^{\text{age}})^2, 0.01, 160]$; without this modifier, $\tilde{k}_{ig} = k_{c_i g}$. The fitted dispersion is an effective observed-variation model; it does not identify all real overdispersion as technical noise. The principal cellular prior has explicit observation dropout disabled. Study-specific detection and structural gene absence are available for their named prior ablations. The general simulator additionally supports composable observation effects, described separately in the supplement.

Real data inform the prior through aggregate calibration targets. The principal ranges were selected to cover bank-matched summaries from ROSMAP, ZEBRA, PBMC CITE-seq and kidney, rather than reproduce one panel's medians. These summaries concern depth, dispersion, donor

variation, abundance concentration, marker structure and separability. The calibrated objects are scalar parameters and distributions; the generator does not load real donor expression profiles or named-gene templates into training banks. Nevertheless, calibration and repeatedly inspected development evaluations used the same real datasets that contribute to the reported evaluations. Synthetic-only model training therefore does not imply a development procedure independent of real data, and the realism comparison is a prior-predictive diagnostic rather than independent validation. Baseline-prior comparisons retain their own configuration and population definitions.

Tensor interfaces, streaming and software extensibility

The software exposes three coordinated interfaces: configurable experiment objects, PyTorch-compatible cell streams[44], and annotated dataset export. A donor-expression provider supplies cell types, gene identifiers and donor-conditioned expression; an external provider can be registered through a Python factory without changing the experiment orchestrator. Generic observation effects implement a tensor-valued forward operation and compose in an owned `EffectStack`; governed stacks validate their topology and terminal count sampler. The cellular-prior provider instead owns its complete observation process, preventing a second application of technical noise through the generic stack.

The common output contract distinguishes normalized expected profiles, realized pre-sampling intensities and observed counts. All are matrices with shape cells $\times$ genes; profiles and intensities use float32 and counts use int32. Their dimensions, non-negativity, integer range and profile normalization are checked at the interface. Public metadata, sampling groups and provenance accompany the tensors, while hidden biological state and training-only sidecars remain separately controlled. This contract prevents a normalized profile from being silently interpreted as observed UMI counts.

`CellStream` is a PyTorch `IterableDataset`. It first reserves cell coordinates and then materializes a microbatch; the generic path groups cells sharing an effect stack before applying tensor operations. The cellular-prior kernel combines NumPy computations[45] with PyTorch tensor interfaces and returns CPU tensors. The current stream implementation uses CPU generation and requires `num_workers=0` in a local PyTorch DataLoader. Process-global Python, NumPy and Torch random states are isolated, activated and restored for each simulator instance. Concurrent generation is provided by independently seeded processes, rather than cloning an active simulator into DataLoader workers.

CellPFN's task producer uses spawned CPU workers with bounded prefetch and deterministic result delivery. Fixed biology banks amortize experiment initialization across multiple support/query views, while prefetch permits CPU production to overlap model consumption. Counts are stored as uint16 only when all values are exactly representable, otherwise retaining int32; there is no count clipping to meet a storage width. Task receipts bind configurations, seeds, shapes and array contents. The model-neutral adaptation bank stores expression views and local labels before model-specific preprocessing, allowing the same cellular tasks to be consumed in different model environments.

For independent simulation use, collators convert streamed output to AnnData[46], preserving count layers, cell/gene annotations and runtime provenance. Public-data and hidden-ground-truth loggers support H5AD, Zarr and Loom outputs. Dense and sparse observed-count exports are available, but the sparse path retains dense expected profiles and intensities. Working memory for direct generation scales with the microbatch's cell-by-gene arrays plus fixed biological state; exported datasets and prefetched banks add their own storage. Thus scalability depends on the selected output contract as well as the sampler. The supplement specifies these memory terms and provides a minimal usage example. Architecture alone is not interpreted as a measured speedup or a completed million-cell benchmark.

Support summaries and same-gene neural relations

The support encoder applies a shared multilayer perceptron independently to each cell–gene pair. Its features include expression, standardized expression and detection, support-wide and class-specific moments, and, for count inputs, square-root composition (full definitions below). The two input projections have widths $10 \to 64$ and $9 \to 64$ for count and processed inputs, respectively, followed by GELU, a $64 \to 64$ linear map and layer normalization. For a sequence of gene features h_g, both the support and relation encoders use the symmetric reduction

$$\mathcal{R}(h) = \left[\frac{1}{G} \sum_g h_g,\ \sqrt{\frac{1}{G} \sum_g h_g^{\odot 2} + \epsilon},\ \max_g h_g,\ \log \sum_g e^{h_g} - \log G \right], \tag{8}$$

with componentwise operations. This retains average, energy, extreme and smoothly aggregated responses while allowing arbitrary gene counts and permutations, in the spirit of set-based representations[32]. The resulting 256-dimensional support-row vector is projected to 448 dimensions, followed by three residual, pre-normalized multilayer perceptron blocks with hidden width 1,792.

Within class c, learned inducing queries assign weights to support rows. If e_i is the row embedding and $\bar{e}_c$ its class mean, summary k has weights

$$a_{cki} = \mathrm{softmax}_{i:y_i=c} \left[\frac{\langle f_q(\bar{e}_c + q_k), f_k(e_i) \rangle}{\sqrt{448}} \right]. \tag{9}$$

Here f_q and f_k are learned affine maps from 448 to 448 dimensions. The implementation allocates eight summary slots and activates $\min(n_c, 4)$, where n_c is class support size; inactive slots are masked. At one support cell per class, precisely one summary is active. Each summary is a weighted average of the per-gene support views, retaining the correspondence between a query gene and that same gene in the support.

For every query q, class c, summary k and gene g, the relation input concatenates the query view, the weighted support view, their elementwise product and absolute difference, and additional support statistics and row-level covariates. It has 28 channels for counts or 22 for processed expression. Separate $28/22 \to 448$ projections feed a shared GELU–linear–layer-normalization network producing

224 relation channels. Applying equation 8 across genes yields an 896-dimensional vector for each query–class–summary combination. A linear projection, GELU and layer normalization map this vector to 448 dimensions, and a learned summary-role embedding is added.

Eight learned program queries cross-attend to the active summary tokens using eight attention heads. The attended output is added to the program queries and layer-normalized. A learned token specifying count or processed mode is prepended, and the resulting nine tokens pass through two pre-normalized transformer layers[47], each with eight heads, width 448, feed-forward width 1,792, GELU and zero dropout. Mean and maximum pooling over the eight transformed program tokens, excluding the mode token, yields an 896-dimensional vector h_{qc}. The direct classifier is a $896 \to 448 \to 1$ multilayer perceptron with GELU, producing one unbounded neural logit per class. Its output contains no fixed analytic score or residual correction to an external classifier. Support-derived analytic quantities are input features only. The backbone without the gene–gene operator has 11,925,345 learned parameters.

Support-fitted gene–gene operator

The operator supplies joint gene information to the neural relation and classification paths. It estimates pooled within-class covariance from the support-standardized vectors in equation 1. Let μ_c be the class mean of these vectors, $r_i = z_i - \mu_{y_i}$, and $\nu = \sum_c \max(n_c - 1, 0)$. A gene is support-variable if its maximum and minimum standardized support values differ. All other genes are masked from the operator, including its distances and channels. On the G_* support-variable genes, for $\nu > 0$, define

$$v_g = \frac{1}{\nu} \sum_i r_{ig}^2, \qquad D_g = v_g + 0.01 \, \frac{1}{\max(G_*, 1)} \sum_{g'} v_{g'}. \tag{10}$$

If the additive floor is at most 10^{-10}, it is replaced by 1; if $\nu = 0$, the diagonal is the identity. These safeguards keep completely constant or within-class constant support finite. Masked genes have zero inverse scale.

We diagonalize the support-row Gram matrix of $RD^{-1/2}/\sqrt{\nu}$ in float64. From its leading eigenvalues η_j and eigenvectors, we recover orthonormal gene loadings U in the diagonally standardized coordinates. The number of candidate components is at most $\min(32, \lfloor \nu/2 \rfloor, N)$; retained spikes are $\lambda_j = \max(\eta_j - 1, 0)$, with zero-spike components discarded. The implied covariance is

$$\widehat{\Sigma} = D^{1/2} \left(I + U \, \mathrm{diag}(\lambda) U^{\mathsf{T}} \right) D^{1/2}. \tag{11}$$

This pooled estimator has no per-class low-rank subspaces. It is fitted without gradients from support cells alone; no trainable covariance matrix or gene-identity embedding is introduced.

For query–class difference $d_{qc} = z_q - \mu_c$, put $a_{qc} = D^{-1/2} d_{qc}$ and $b_{qc} = U^{\mathsf{T}} a_{qc}$. The squared

Mahalanobis distance and whitened difference are

$$M_{qc} = \|a_{qc}\|^2 - \sum_j \frac{\lambda_j}{1+\lambda_j} b_{qcj}^2, \tag{12}$$

$$w_{qc} = \Big[I - U\,\mathrm{diag}\Big(1 - (1+\lambda)^{-1/2}\Big)\,U^{\mathsf{T}}\Big]\,a_{qc}. \tag{13}$$

Before clipping, $\|w_{qc}\|^2 = M_{qc}$; computed distances are lower-bounded by zero to remove round-off artifacts. The gate

$$\gamma_c = \begin{cases} (n_c - 1)/(n_c + 3), & n_c \geq 2, \\ 0, & n_c \leq 1 \end{cases} \tag{14}$$

sets the operator's contribution to exactly zero for one-shot classes. The gene channel is $\gamma_c\,\mathrm{clip}(w_{qcg}, -8, 8)/4$. A bias-free $1 \to 448$ projection adds this channel to the relation network's input projection, shared across summaries.

The second path provides class-comparative operator information to the classifier. With $\mathcal{C}_+ = \{c : n_c \geq 2\}$, define

$$t_{qc} = \frac{M_{qc} - |\mathcal{C}_+|^{-1} \sum_{c' \in \mathcal{C}_+} M_{qc'}}{\sqrt{2\max(G_*, 1)}}, \tag{15}$$

$$s_{qc} = \mathrm{asinh}(t_{qc}) - \frac{\sum_{c'} \gamma_{c'}\,\mathrm{asinh}(t_{qc'})}{\sum_{c'} \gamma_{c'}}. \tag{16}$$

Empty-set denominators are replaced by 1; all contributions then vanish through the gate. If r is the retained pooled rank, the three inputs are $u_{qc} = (\gamma_c s_{qc}, \gamma_c r/32, \gamma_c)$. A $3 \to 448 \to 896$ GELU multilayer perceptron f modifies the classifier input as $h_{qc} + \gamma_c f(u_{qc})$. Thus both the scalar inputs and the head output are gated. The second centering in equation 16 ensures $\sum_c \gamma_c s_{qc} = 0$ even for unequal support sizes. The gene projection and the output layer of f are initialized to zero, preserving the common parent model's predictions at initialization. These paths add 404,544 parameters, giving 12,329,889 parameters in total.

We distinguish three trained arms: no operator, diagonal operator and full operator. The diagonal arm uses equations 10–16 with no low-rank correction, $\widehat{\Sigma} = D$, while retaining the full support fit's rank scalar $r/32$. Its neural dimensions, gate and initialization match the full arm. The no-operator arm omits both added neural paths. All three arms continue from the same no-operator parent with matched task schedules. Separately, operator-off and rank-zero interventions remove pathways or low-rank corrections during inference with trained parameters fixed. These interventions measure dependence of an existing predictor on its operator; they are not substitutes for training the corresponding ablation. One-shot gating likewise does not require independently trained arms to agree, because their shared backbone parameters can diverge during training.

Reusable simulated banks, task sampling and optimization

The trainer receives simulated counts from pyScIntelligentDesign through a deterministic bank interface. Each independently seeded bank contains eight donors, eight cell types and 256 cells per donor–type combination at 8,192 parent genes. Thirty-two episodes reuse each bank with separately sampled targets, cells, gene panels, gene orders and class-label permutations. Four banks are interleaved, with one episode per bank in each four-task optimizer opportunity. Reuse therefore increases task exposure without creating additional independent biological datasets. During cellular-prior bank construction, a row with fewer than twenty counts receives up to eight technical redraws at the same biological coordinates. The highest-count attempt is retained, stopping when the threshold is reached; any remaining zero-count row invalidates the bank.

Targets comprise binary donor metadata, four-category donor metadata, cell type and donor identity. Metadata targets use one query donor per label and other eligible donors as support, within one selected cell type. Cell-type targets use two query donors and six support donors. Donor-identity tasks use the same donor classes but disjoint support and query cells. They consequently train identification, not transfer of a phenotype to unseen donors. The first 32 cells of every donor–type combination are reserved for diagnostics. Support cells are sampled by cycling across eligible donors. Metadata values are assigned native binary labels or rank-based binary/four-category labels without splitting ties; incompatible target fields are replaced by other eligible fields, and a bank with no compatible target produces a recorded zero-weight task.

Support budgets are drawn from 1, 10 and 100 cells per class with probability 0.25 each, 2, 4, 8 and 16 with probability 0.04 each, and 32 and 64 with probability 0.045 each. Requested gene panels contain 2,048, 4,096 or 8,192 genes, sampled without replacement and permuted jointly for support and query. A 12.5-million-element cap reduces gene width when necessary while preserving requested cell counts. The nominal query budget is 100 cells per class; in the cellular prior's cell-type tasks, the total $100C$ is distributed according to available class frequencies. The curriculum laws, target allocation and admission rules are specified in Supplementary Table S13.

The evaluated model was trained on 320,000 scheduled episodes from 10,000 baseline-prior banks, followed by 64,000 episodes from 2,000 banks of the cellular prior. The gene–gene paths were added at the start of this 64,000-episode stage. There were 96,000 optimizer opportunities in total; 276,828 episodes contributed nonzero loss, including all 64,000 final-stage episodes. These are distinct from the number of banks or simulated cells. All CellPFN arms use model seed 42 and task-stream seed 15113. The endpoint was the declared training milestone, rather than a checkpoint selected by maximum real-panel accuracy. Real summaries informed simulator calibration, and development panels informed design decisions; synthetic-only optimization does not imply that the complete research process was independent of real data.

The objective is mean query cross-entropy within each task, averaged uniformly over accepted tasks in an optimizer opportunity. Rejected tasks contribute zero weight; an opportunity with no accepted task does not apply an optimizer step. AdamW[48] uses $\beta_1 = 0.9$, $\beta_2 = 0.999$ and weight decay 10^{-4}. Its learning-rate multiplier has 8,000 warm-up opportunities and cosine decay to a 0.1

floor over a 200,000-opportunity horizon. The base rate is 10^{-4} for the first 40,000 opportunities and 10^{-5} thereafter. At the operator stage, pre-existing Adam moments and counters are restored by parameter name; new operator parameters start with fresh optimizer state and use ten times the backbone rate, multiplied by a linear 1,000-opportunity ramp. Backbone and operator parameter groups are separately clipped to gradient norm 5. Training uses PyTorch[44] with bfloat16 autocast on NVIDIA H100 80-GB GPUs; feature statistics and losses use float32 and the covariance fit uses float64. Gene and query–class chunking, and activation checkpointing during training, bound working memory without changing the mathematical reduction.

Continued training of tabular foundation models

We initialized TabICL and the small and large Kumo Tabular models from their released weights[15,36]. The three adaptation arms consumed the same cellular task stream and generic replay tables. Each of 2,500 updates contained 16 cellular tasks and four generic tables: 40,000 cellular tasks and 10,000 replay tasks. Cellular tasks retained the simulator's target construction, used 1, 10 or 100 support cells per class in exact thirds over consecutive triples of tasks, and used 32 query cells per class. Task families followed the final curriculum probabilities (60% binary metadata, 20% categorical metadata, 15% cell type and 5% donor identity), with no more than eight classes.

Cellular counts were log-normalized and transformed using support-fitted 2k HVGs or a principal-component view with at most 100 components. View choice, preprocessing ensemble choice and randomness were recorded per task. Neither feature selection, standardization nor projection was fitted to query rows. TabICL used its support-fitted ensemble generator, with identity or power normalization, Latin feature shuffling, shifted class labels and outlier threshold 4. Training used one or two ensemble members per cellular task, averaging their query cross-entropies. Kumo used its own default preprocessing recipe fitted to support rows: four recipe members expose its four numerical transformations, and the task plan selects one or two members for gradient accumulation. Gradients pass through the classifier behind Kumo's class-count wrapper. The module remains in evaluation mode during this differentiable route, which has no dropout or batch-statistic update.

Generic replay tables were drawn from TabICL's mixed structural-causal-model prior with 5–100 features, at most ten classes, at most 1,024 rows and support fractions between 0.1 and 0.9. For Kumo, replay query rows whose class is absent from support are masked. The update loss averages valid task losses. All three principal adaptation arms used float32 training, AdamW with peak learning rate 3×10^{-5}, zero weight decay, $\beta = (0.9, 0.999)$, 125 linear warm-up updates, cosine decay to zero at update 2,500 and global norm clipping at 1. Recorded invalid tasks were bounded by a cap of 40, rather than silently replaced. Checkpoints at updates 313, 625, 1,250 and 2,500 correspond to approximately 5,000 and exactly 10,000, 20,000 and 40,000 cellular tasks. Evaluation used each model's deployment preprocessing; Kumo used eight ensemble members for the small model and 16 for the large model, with seed 0. No additional supervised head was fitted for these in-context predictions.

The generic-only Kumo control replaced the 16 cellular slots with generic tables and retained four

replay slots; its row and column counts were not matched to the cellular stream. Two further cellular task draws retained the same released initialization and model seed, quantifying task-sampling variation rather than independent initialization variation. Prior-content proxy comparisons used the same adaptation machinery and a declared 10,000-cellular-task endpoint on the 2,500-update schedule. Changes in task family, donor count or class composition are reported as interventions in that distribution, not architectural changes.

Biological datasets and support–query separation

PBMC evaluation used the 3-prime RNA component of the Hao et al. CITE-seq reference[33]: GEO accession GSE164378, RNA sample GSM5008737 and the supplied cell metadata. The source contains 161,764 cells and 33,538 RNA coordinates. Published annotations combine RNA, measured surface proteins and curation; they are not an independent protein-only ground truth. Predictions used RNA alone. Barcodes joined the original RNA matrix to metadata, and RNA library sums were checked against the metadata. The accompanying antibody measurements and annotations were audited against the author-linked reproduction snapshot; antibody measurements were not prediction inputs. Cells annotated as doublets were excluded. A class was eligible only when it contained at least 25 cells in each support donor and ten in each query donor, yielding eight broad and twenty finer classes.

Support donors were P1, P3, P5 and P7; query donors were P2, P4, P6 and P8. All timepoints of a donor remained on the same side, with two donors from each acquisition batch on each side. Ten query cells per class and donor were selected by deterministic metadata-based SHA256 rankings, giving 320 and 800 query cells for the broad and fine tasks. Within each class and support draw, donor and cell rankings were shuffled deterministically and sampled round-robin across donors. Prefixes supplied one, ten or 100 support cells per class; at 100 supports, each donor contributed 25 cells. Query cells were fixed across support budgets, draws and methods. The primary comparison used support draws 0–2; extended diagnostics followed the same sampling rule with more draws.

Human cortex evaluation used the 50,000-cell ZEBRA benchmark subsample[34], with 21,158 genes and four broad classes: glia, immune, neural and vascular cells. The shared support pool comprised Bakken, Gerrits, Nagy, Schirmer, Trobisch and Yang Cortex studies. Development queries comprised 7,769 cells from 59 recorded donors in Absinta, Kihara, Lim and Velmeshev studies; held-out-study queries comprised 7,163 cells from 34 recorded donors in Hardwick, Herring, Jaekel and Yang studies. Support and query studies were disjoint, and query cells remained fixed across budgets and methods. Both query panels nevertheless contributed to model development; the held-out-study designation describes their study separation, not an untouched final model-selection set. Frozen cell identifiers, study assignments and input-file hashes define these panels more precisely than publication names alone.

The secondary kidney evaluation used healthy living-donor data from GSE202109[49], distributed through CELLxGENE, with 27,675 cells from 19 donors and 26,791 measured genes. Three donor-held-out folds evaluated five- and nine-class cell-type tasks and proximal-tubule sex prediction.

Cell-type tasks supplied 100 supports per class and 40 query cells per class. Sex prediction used six or seven query donors per fold, with five query cells per donor. Two 4,096-gene panels were retained: a kidney-support-derived panel and a panel derived from the historical ZEBRA support pool. These feature panels therefore used more support information than an individual few-shot case. Kidney cell-type accuracy and donor-level sex accuracy were kept separate.

Expression inputs and feature selection

CellPFN received original counts for PBMC and the supplied processed expression for the primary cortex analysis. Count-based comparator inputs were transformed to $\log[1 + 10^4 x_{ig} / \sum_h x_{ih}]$, using each cell's full measured library before gene selection. Supplied ZEBRA log-transformed counts per 10,000 were passed through unchanged. Genes unavailable in individual ZEBRA studies had been stored as zeros; this preprocessing artifact was evaluated separately from biological non-expression.

Highly variable genes were selected using labelled support rows only. The implementation exponentiated log-expression, computed gene means and unbiased variances, and standardized log dispersion within 20 mean-expression bins, following the Seurat-flavour dispersion statistic. Finite scores were ranked with a deterministic tie break based on the anonymous support column, retaining at most 2,000 or 8,192 genes. The same selected columns and order were applied to queries. Table labels denote requested HVG caps. In the three primary one-shot draws, the 8,192-HVG view retained 6,995–7,647 genes for broad PBMC and 2,209–5,301 for cortex; finite support variation could therefore reduce the realized width below the requested cap. Principal-component controls standardized support-selected genes, clipped standardized values to $[-10, 10]$, fitted the decomposition to support rows, and retained $\min(100, n_{\text{support}} - 1, n_{\text{genes}})$ components. Query expression did not influence selection, scaling or projection.

The cortex sensitivity analysis reconstructed integer counts from the lattice of the supplied processed values and retained 12,325 genes measured across eligible contributing studies. Reconstruction was conditional on the smallest positive expression level representing one count, rather than a retrieval of original sequencing counts. A matched processed partner was recomputed from the reconstructed common panel. Query identities and labels were unchanged; some supports changed after reconstruction-ineligible cells were removed. Consequently, historical full-panel and common-panel results do not isolate count versus processed representation. The reconstruction rule and support replacement are specified in the technical supplement.

Comparator models and selection

Released tabular comparators were TabICLv2, TabPFN-3.5 and Kumo Tabular small and large [15,36,37]. Each used labelled support cells as context, with no target-task gradient update. Continued-training checkpoints used the same inference route as their released reference. Main comparisons used support-selected 2,000-gene inputs; scaling analyses separately evaluated 8,192-gene and all-gene inputs. Package-default transformations, feature subsampling and feature limits remained active. Refused inputs and unevaluated cases were recorded as missing, not as zero accuracy. TabPFN-3.5

results were obtained under its non-commercial research terms[38].

Geneformer-V2-316M[35] and the whole-human scGPT checkpoint[3] supplied frozen embeddings. Geneformer used Ensembl-linked rank-value tokens and mean embeddings; scGPT used gene-symbol inputs, binned values and unit-length CLS embeddings. Logistic classifiers were fitted to support embeddings only. Geneformer used support-standardized embeddings; scGPT used unscaled embeddings. Exact token limits, classifier settings and recorded software releases are listed in Table S14. Native input routes were preserved, so these comparisons jointly reflect representation, feature coverage and classifier choice.

Support-trained comparators included nearest-class-mean and prototype rules, logistic regression, diagonal and low-rank LDA, CellTypist[50], SingleR[51] and scANVI[52]. CellTypist was fitted to the selected supports without query majority voting. SingleR used the supports as its reference. Inductive scANVI used supports only; its separately labelled transductive comparison also used unlabelled query cells and a donor batch key. Processed cortex inputs were not passed to count-likelihood scANVI. LDA diagnostic rows were available at ten and 100 supports, not one support per class. The inductive scANVI recipe used batch size 128, a 90% training fraction and ten supervised epochs without class-balanced replacement sampling. At fine-PBMC one-shot support, 18 of 20 labelled cells entered optimization, leaving at least two classes without a training example. These low-support results describe that fixed recipe. SingleR retained its default classic marker mode rather than the alternative recommended for single-cell references.

The displayed family representatives were chosen within the development evaluation. Kumo size selection used the mean across the twelve core panel–budget combinations on 2,000 genes; the CellPFN reference was an explicitly selected model state, and logistic heads were the declared primary heads for Geneformer and scGPT. These choices were not independently validated model-selection procedures. Study-level pretraining overlap was assessed separately from benchmark support–query separation, using the recorded corpus/source-study audit. Such matches indicate possible unlabelled query-source exposure; they do not establish identical cell inclusion, target-label exposure or supervised leakage.

Exploratory subtype and sex tasks

The eight PBMC subtype tasks comprised level-2 CD4 and combined T-cell siblings, level-3 CD4, CD8, B, NK and combined T-cell siblings, and the complete eligible level-3 vocabulary. They inherited the PBMC donor split and class-availability rules. Candidate selection was conducted after the prior-ablation read-outs and is exploratory. It used nearest-class-mean and logistic-reference performance, not CellPFN or proxy predictions: at 100 supports, the larger reference accuracy had to reach a chance-corrected skill of 0.20 while remaining below 85% balanced accuracy. Each task used ten support draws at one and ten supports and five draws at 100 supports. CellPFN received up to 8,192 support-selected genes; the Kumo-small proxy received up to 2,000. Supplied CellPFN task means were reported without constructing intervals from unavailable individual-case predictions. Two exploratory cortex tasks used the same reference checkpoint and up to 8,192 HVGs: seven

development labels and three held-study neural labels, defined in Supplementary Table S9. They used ten, ten and five support draws at one, ten and 100 supports; the supplied checkpoint means do not include draw-level uncertainty. Changes in labels and evaluated cells preclude interpreting their difference from the primary broad-class benchmark as an isolated difficulty effect.

The original ZEBRA sex task used Sayed oligodendrocytes, 100 support cells per sex from twelve GSM samples, and 50 query cells from ten other GSM samples. GSM separation was not interpreted as independently verified donor separation. Later study-matched sex tasks used ten splits, four to 32 support units per sex, ten cells per support unit and twenty per query unit. The positive control required measurement of XIST and at least four specified Y-linked genes; a broader task included additional studies with a different common gene universe. Units were persons except for age-by-sex groups of GSM samples in Sayed. Reused persons and unresolved cross-study identity overlap preclude treating the ten splits as independent cohorts. Scores were averaged within query units before unit-level AUC calculation. **Gene-level attribution for these sex tasks remains unavailable; eligibility based on sex-linked genes does not demonstrate that those genes drove the model's predictions.**[O06]

Metrics and uncertainty

Balanced accuracy was the mean recall across every declared query class. Negative log-likelihood was the mean negative log probability of the true class, in nats; expected calibration error used fifteen fixed-width confidence bins. Probabilities and class order were checked before scoring. Performance means first averaged support draws within a panel–budget combination and then weighted the twelve combinations equally. Low-support summaries averaged the eight combinations with one or ten supports. Paired comparisons matched model, panel, input view, support budget and draw; a difference between unmatched views was not interpreted as an isolated model effect.

Three-draw benchmarks are descriptive and use ranges or individual values. The ten- and twenty-draw diagnostics used Student t intervals over support draws, paired for contrasts. These quantify sensitivity to labelled-context sampling conditional on the chosen cohorts and weights. Every trained arm had one model seed; task-stream replicates did not estimate variation across independently initialized models. Kidney fold differences combine cohort and support variation. The nested PBMC subtype tasks share cells and labels and are not independent biological replicates. Attribution case ranges and single-run timings were not assigned confidence intervals. Missing, refused and unrun values were never replaced by zero.

Query-gene deletion and attribution validation

For a query x and fixed support set S, let a and b be the initially highest- and second-highest-logit classes. Define the fixed-pair margin $m(x; S) = f_a(x; S) - f_b(x; S)$. The exact deletion target for gene g was

$$I_g(x; S) = m(x; S) - m(x^{(g \leftarrow 0)}; S). \tag{17}$$

Only the query gene was set to zero; support summaries and the fitted covariance operator were held fixed. A positive effect supports the original predicted class relative to its original runner-up. It is neither an expression-direction statistic nor an experimental perturbation effect. Negative effects oppose this fixed-pair margin. A gene already at zero has zero deletion effect, so this intervention cannot measure predictive information carried by its absence.

DS-LOO approximates all query-gene deletions by updating the mean, root-mean-square, maximum and log-sum-exp pools when each gene's relation vector is replaced by its zero-expression counterpart. It also updates the gene–gene operator's class-distance scalars and reevaluates their learned pair embedding. A first-order expansion of the prediction head combines the pooled and pair-embedding changes. The own-gene whitening-channel change is retained, whereas changes propagated to other genes through off-diagonal whitening are omitted. An additional gradient calculation supplies first-order corrections for query read depth in count mode and for the detected-gene fraction. This is a gene-attribution sweep with head differentiation and normalization corrections; it is neither an ordinary prediction pass nor exact finite deletion. The technical supplement specifies the estimator, numerical stabilization and validation boundaries.

Validation used six case configurations and both the reference and no-operator model: three PBMC cases, one cortex case, and one kidney case evaluated in count and processed modes. Cells had to be correctly classified under both stored bfloat16 predictions and the float32 rerun. Seeded sampling selected at most 24 cells per class, or 40 for kidney. Exact deletion was evaluated on unions of method-selected genes and random expressed genes, with additional DS-LOO extreme-ranked genes; it was not exhaustive over all genes. Rank correlations were computed within cells, then summarized by the recorded class/case aggregation; relative Euclidean error compared estimated and exact effect vectors. Deletion curves recomputed the fixed-pair margin after removing sets of ranked query genes. Gene symbols were restored after prediction for descriptive marker comparisons. No marker enrichment test was performed. Timings distinguished measured attribution runs from extrapolated exhaustive-deletion costs; comparisons made under shared GPU load remain provisional.

Provisional neuropathology analysis

The ROSMAP analysis used aggregate records for donor-disjoint splits of neuropathology contrasts[42]: CERAD 4 versus 1, CERAD 3–4 versus 1–2, and Braak 0–II versus V–VI. Support and query allocation was matched within sex on source and age bin. Each support donor contributed eighteen cells and each query donor twenty. Donor-level AUC was summarized over ten overlapping random splits using the reported corrected repeated-split interval. The exact sequencing-release crosswalk, donor-label corrections and CellPFN input-mode record remain unresolved, and these analyses must be rerun after cohort definitions are frozen.[O02]

Biological visualization and figure provenance

RNA maps (Fig. 3; Supplementary Fig. S7) were computed from the exact support/query unions of four frozen cases: coarse PBMC at 100 supports per class, fine PBMC at ten, and development and held-study cortex at ten, all with support draw zero. These comprise 320, 800, 7,769 and 7,163 query cells, respectively. Prediction labels and probabilities were recovered from the reference checkpoint's cached evaluation bundles, with matching observation identifiers, gene order and class order. Model predictions were not recomputed or changed by visualization.

For count inputs, rows were normalized to 10,000 counts and log1p-transformed; already processed cortex values were retained. Using Scanpy[53], a visualization-only Seurat-flavour selection retained up to 2,000 variable genes from the unlabelled support/query union. Genes were standardized and clipped at ten standard deviations. Principal-component analysis retained at most 50 components, followed by a 15-neighbour Euclidean graph and UMAP[54] with minimum distance 0.3 and random seed 42. Coordinates were fitted once per case without annotation or prediction labels, then reused for reference, prediction, error, donor/study and gene overlays. Only query cells are displayed. These coordinates are RNA visualizations and were never used as classifier features.

The expression and detection matrices in Fig. 3c summarize labelled supports; spatial expression overlays in Supplementary Fig. S7 show query values. Attribution panels use the original selected correctly classified query cells. Case-local attribution indices were joined to query barcodes through the original case manifests, with gene symbols restored in benchmark column order. Unexplained cells remain visibly distinct from zero-valued explanations. Expression and signed-attribution colors span the observed ranges independently; attribution colors are centered at zero. No cells or effect outliers were removed for visual appearance. The synthetic illustration uses one predetermined mean-program bank from the stated cellular configuration and displays all 16,384 generated cells; its anonymous genes and embedding are separate from the real-data maps. This illustration is not an additional realism or throughput measurement.

The supplementary sex/age display (Supplementary Fig. S6) preserves the existing exploratory reference-checkpoint tasks. Sayed and Gerrits oligodendrocyte age targets use the study median among eligible units. Herring excitatory-neuron development uses four age classes (< 1, $1–< 6$, $6–< 16$ and ≥ 16 years), three supports per class and twelve query persons; its AUC is macro one-versus-rest. Split-level means and ranges remain descriptive because units are reused. Recorded computational-cost panels (Supplementary Fig. S9) distinguish adaptation optimizer compute, measured attribution time and extrapolated or estimated alternatives; they are not controlled end-to-end speed comparisons.

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Supporting quantitative results

These tables accompany the main Results and retain negative findings, exploratory analyses and limits on replication. Unless stated otherwise, intervals describe variation over support draws for fixed models and query panels; they do not represent independent model training or population-level biological replication.

Table S1: Panel-specific benchmark performance and contextual baselines.

Model	1 support/class	10 supports/class	100 supports/class
PBMC coarse			
CellPFN	60.1 [52.5, 70.9]	84.0 [81.9, 85.9]	91.9 [91.6, 92.2]
TabICLv2	50.2 [41.3, 58.4]	81.4 [79.4, 83.1]	92.4 [91.3, 93.4]
TabICLv2 + cellular prior	62.3 [48.1, 72.5]	**85.2** [81.9, 88.1]	**93.4** [93.1, 94.1]
Kumo small	45.8 [40.3, 50.3]	82.2 [80.6, 83.4]	92.6 [91.6, 94.7]
Kumo small + cellular prior	**62.8** [52.8, 69.7]	83.3 [80.9, 86.3]	92.3 [91.6, 93.1]
Kumo large	58.9 [46.3, 65.6]	81.7 [78.8, 83.8]	92.8 [91.9, 93.8]
Kumo large + cellular prior	62.2 [50.3, 70.0]	82.5 [79.4, 85.0]	92.8 [91.9, 93.4]
TabPFN-3.5	34.3 [30.9, 38.4]	82.6 [80.6, 83.8]	91.5 [90.0, 92.5]
Geneformer-V2 + LR	**67.1*** [58.1, 74.7]	**88.5*** [86.9, 90.3]	92.7* [91.6, 93.4]
scGPT + LR	64.9* [53.1, 72.2]	77.5* [75.6, 78.8]	80.7* [80.3, 81.3]
Low-rank LDA	–	79.4 [78.8, 80.0]	90.0 [89.7, 90.3]
CellTypist	39.8 [33.8, 47.5]	77.5 [76.9, 78.1]	87.8 [87.5, 88.1]
SingleR	56.7 [52.8, 60.9]	78.2 [74.1, 80.9]	85.9 [85.0, 86.6]
scANVI inductive	20.0 [11.3, 32.5]	40.7 [26.6, 51.3]	86.7 [86.6, 86.9]
scANVI transductive	40.5 [36.3, 44.1]	69.1 [65.9, 74.7]	89.1 [88.1, 90.9]
Logistic regression	37.5 [34.4, 43.4]	80.5 [79.1, 81.6]	91.7 [90.6, 93.1]
PBMC fine			
CellPFN	**51.1** [44.8, 55.4]	79.0 [77.6, 80.4]	89.0 [88.6, 89.1]
TabICLv2	30.0 [26.8, 34.3]	73.8 [72.1, 75.6]	83.6 [82.9, 84.5]
TabICLv2 + cellular prior	29.1 [26.4, 32.3]	64.6 [61.0, 67.9]	85.3 [84.3, 86.1]
Kumo small	44.1 [43.6, 44.6]	80.0 [79.4, 80.4]	88.3 [88.1, 88.4]
Kumo small + cellular prior	49.0 [45.9, 52.3]	80.5 [80.0, 81.0]	88.7 [87.6, 89.6]
Kumo large	46.3 [42.3, 50.0]	**83.3** [82.8, 83.6]	**89.3** [89.1, 89.5]
Kumo large + cellular prior	50.9 [47.0, 55.8]	82.2 [81.4, 83.5]	89.2 [89.1, 89.4]
TabPFN-3.5	33.0 [29.4, 36.5]	79.3 [77.4, 80.4]	86.9 [86.5, 87.3]
Geneformer-V2 + LR	**68.8*** [66.6, 70.9]	**84.9*** [84.3, 85.5]	**90.4*** [90.0, 90.9]
scGPT + LR	55.7* [51.3, 58.1]	65.9* [65.3, 67.1]	72.0* [71.5, 72.9]
Low-rank LDA	–	70.3 [70.0, 70.5]	87.8 [87.5, 87.9]
CellTypist	33.9 [26.8, 37.8]	69.7 [68.6, 70.8]	84.2 [83.9, 84.6]
SingleR	46.0 [42.1, 51.8]	68.6 [67.0, 70.1]	83.3 [82.6, 83.9]
scANVI inductive	21.7 [14.4, 26.1]	51.8 [50.8, 53.4]	83.9 [81.6, 85.3]
scANVI transductive	40.8 [39.5, 41.5]	70.8 [68.8, 74.0]	86.2 [85.8, 86.5]
Logistic regression	39.0 [34.0, 46.8]	73.4 [71.8, 75.9]	85.1 [84.4, 85.6]
ZEBRA development			
CellPFN	41.4 [33.1, 47.4]	**63.6** [61.9, 65.6]	**66.1** [65.8, 66.4]
TabICLv2	39.5 [33.4, 42.6]	59.9 [57.9, 62.2]	65.3 [64.9, 65.7]
TabICLv2 + cellular prior	44.6 [35.9, 57.1]	60.6 [57.4, 62.6]	65.3 [65.1, 65.5]
Kumo small	31.2 [26.6, 36.4]	57.5 [50.5, 61.7]	65.1 [64.5, 65.7]
Kumo small + cellular prior	41.9 [34.9, 51.2]	59.2 [53.4, 62.3]	64.9 [63.8, 65.6]
Kumo large	40.1 [38.2, 41.5]	56.8 [49.2, 61.2]	65.1 [64.9, 65.3]
Kumo large + cellular prior	42.8 [37.3, 52.3]	60.6 [55.6, 63.4]	65.8 [65.6, 66.1]
TabPFN-3.5	38.4 [32.8, 43.3]	59.3 [56.6, 60.8]	65.5 [64.8, 66.0]

Table S1 (continued)

Model	1 support/class	10 supports/class	100 supports/class
Geneformer-V2 + LR	43.1$^{(*)}$ [34.8, 50.2]	61.1$^{(*)}$ [55.3, 66.7]	65.4$^{(*)}$ [65.3, 65.5]
scGPT + LR	**47.6** [43.7, 55.0]	60.2 [53.2, 66.5]	65.8 [65.6, 66.1]
Low-rank LDA	–	60.6 [58.4, 62.0]	66.0 [65.8, 66.4]
CellTypist	34.9 [32.6, 37.3]	57.8 [48.6, 64.0]	63.9 [63.5, 64.6]
SingleR	44.4 [34.4, 53.3]	61.7 [59.8, 63.5]	63.9 [63.2, 64.2]
scANVI inductive	–	–	–
scANVI transductive	–	–	–
Logistic regression	34.7 [28.1, 39.6]	57.4 [48.7, 62.4]	65.5 [65.1, 65.8]
ZEBRA held-study			
CellPFN	62.8 [45.2, 77.1]	**91.9** [88.6, 94.7]	**94.7** [94.3, 95.0]
TabICLv2	49.8 [46.6, 55.2]	83.2 [75.5, 88.1]	92.9 [92.1, 93.5]
TabICLv2 + cellular prior	62.7 [37.8, 81.0]	85.8 [81.9, 88.7]	93.2 [92.2, 94.0]
Kumo small	48.8 [31.7, 65.9]	84.8 [81.3, 86.7]	93.7 [92.3, 94.7]
Kumo small + cellular prior	59.3 [41.8, 73.5]	86.8 [83.4, 88.7]	92.9 [91.1, 94.4]
Kumo large	49.1 [39.6, 56.6]	85.1 [80.3, 88.8]	93.1 [91.7, 94.5]
Kumo large + cellular prior	60.8 [39.1, 75.8]	85.9 [81.4, 88.6]	93.4 [93.2, 93.6]
TabPFN-3.5	54.1 [48.0, 65.3]	85.3 [84.9, 85.8]	93.9 [92.5, 95.3]
Geneformer-V2 + LR	58.4* [44.2, 72.5]	87.4* [83.8, 91.7]	90.3* [88.4, 92.6]
scGPT + LR	**65.5** [52.5, 80.7]	87.2 [80.6, 93.1]	89.5 [88.9, 90.2]
Low-rank LDA	–	91.0 [89.6, 93.8]	**94.7** [94.0, 95.4]
CellTypist	44.4 [41.6, 47.5]	87.8 [80.4, 92.5]	91.4 [90.6, 92.3]
SingleR	56.8 [38.4, 71.1]	86.6 [79.7, 90.8]	89.1 [86.8, 90.6]
scANVI inductive	–	–	–
scANVI transductive	–	–	–
Logistic regression	45.0 [42.1, 48.1]	73.5 [70.7, 76.0]	90.6 [90.0, 91.5]

Values are mean [minimum, maximum] balanced accuracy (%) over the same three support draws, not confidence intervals. Within each panel and support budget, bold identifies the highest mean overall and among unflagged comparators, including ties at displayed precision. Gray means and ranges carry the exposure stars defined in Table 1; unflagged entries do not certify absence of exposure. Inputs follow Table 1; CellTypist, SingleR and low-rank LDA use all genes, whereas logistic regression and scANVI use 2k HVGs, a requested cap of up to 2,000 support-selected genes that can yield fewer genes (Methods). CellTypist [50] is fitted to the support cells. SingleR [51] uses support cells as its reference and the default classic marker method; the exported source notes that the single-cell-reference vignette recommends a different marker setting, so these scores describe the declared default route. The scANVI [52] recipe uses 400 scVI epochs followed by ten scANVI epochs, batch size 128, training fraction 0.9 and no per-label resampling (`n_samples_per_label=None`). In fine-PBMC one-shot inductive training, only eighteen of twenty labelled support cells enter training, leaving at least two classes without a labelled training example. Low-support values therefore describe this constrained recipe. The transductive route additionally uses the unlabelled query pool and is contextual rather than an inductive comparison. Dashes are missing evaluations, never zero accuracy. All selected fitted model checkpoints have one training seed; support-draw variation does not measure training uncertainty. Geneformer-V2 uses the 316M checkpoint. Source-study corpus overlap does not establish exact-cell or target-label exposure. Native gene vocabularies and training histories differ. Family variants were selected using development evaluations; this is not a newly frozen independent benchmark.

Table S2: Training exposure, prior transfer and generic-task retention.

Model	Quantity	Count
CellPFN	Parameters	12329889
CellPFN	Optimizer updates	96000
CellPFN	Simulated banks	12000
CellPFN	Scheduled episodes	384000
CellPFN	Loss-bearing episodes	276828
TabICLv2	Parameters	27552258
TabICLv2	Additional cellular tasks	40000
TabICLv2	Adaptation updates	2500
Kumo small	Parameters	27458266
Kumo small	Additional cellular tasks	40000
Kumo small	Adaptation updates	2500
Kumo large	Parameters	213668250
Kumo large	Additional cellular tasks	40000
Kumo large	Adaptation updates	2500

CellPFN banks are reused for multiple episodes; scheduled and loss-bearing episodes are distinct. Adaptation uses 40,000 cellular tasks and 10,000 generic replay tasks per arm. Released-model pretraining precedes this adaptation and is not included in these counts. TabICLv2 reports 35 million synthetic pretraining datasets[15]; this published count is distinct from CellPFN banks and scheduled episodes. These units do not establish a matched-compute comparison. **Adapted minus released balanced accuracy (percentage points).**

Model	1 support	10 supports	100 supports
TabICLv2	+7.30	-0.50	+0.77
Kumo small	+10.79	+1.31	-0.22
Kumo large	+5.58	+1.08	+0.26

Each delta uses released weights re-scored in the same matched scoring run as the adapted checkpoint. Table 1 instead reports the stored benchmark scores for released models; subtracting its displayed means need not reproduce these paired deltas.

Kumo-small low-support change relative to released weights.

Training stream	Mean change (percentage points)
Cellular task stream 1	+6.05
Cellular task stream 2	+5.73
Cellular task stream 3	+5.17
Generic-table control	-0.05

The low-support average weights eight panel–budget combinations equally (three support draws each). The three cellular runs vary the task stream with the same model seed. The generic control changes rows and columns per update, so this is not a matched-data or matched-cost contrast. No corresponding generic-only control was run for TabICLv2 or Kumo large.

Retention on four generic classification datasets.

Model	One-shot BA	One-shot accuracy	All-budget BA	All-budget accuracy
TabICLv2 released	70.9	69.3	87.8	87.2
TabICLv2 adapted	64.0	62.0	84.8	84.3

Balanced accuracy (BA) and ordinary accuracy (%) equally weight breast cancer, digits, iris and wine, with five split seeds each. The all-budget mean additionally weights 1, 10 and the full training split equally. These exploratory retention checks do not estimate universal tabular performance.

Table S3: Gene-panel scaling and random-gene controls.

Panel	Support/class	All minus 2k HVGs (pp; 95% CI)
PBMC coarse	1	-3.31 [-5.20, -1.42]
PBMC coarse	10	+0.91 [-0.11, 1.92]
PBMC coarse	100	+1.31 [0.92, 1.71]
PBMC fine	1	-1.96 [-3.27, -0.65]
PBMC fine	10	-0.55 [-1.25, 0.15]
PBMC fine	100	+1.81 [1.44, 2.18]
ZEBRA development	1	+0.26 [-2.58, 3.10]
ZEBRA development	10	+2.52 [1.66, 3.39]
ZEBRA development	100	+1.30 [0.98, 1.61]
ZEBRA held-study	1	+1.11 [-2.05, 4.27]
ZEBRA held-study	10	+7.41 [5.26, 9.57]
ZEBRA held-study	100	+4.38 [3.69, 5.07]

Paired differences in balanced accuracy (percentage points) use twenty support draws per panel–budget combination. Intervals are the exported paired 95% t intervals over support draws. They condition on the evaluated checkpoint and query set; they do not cover training, donor-population or panel-selection uncertainty. **Mean balanced accuracy across the twelve panel–budget combinations.**

Gene panel	Mean balanced accuracy (%)
Random 2,000 genes	62.9
2k HVGs	72.4
Random 8,192 genes	69.4
8,192 HVGs	**74.1**
All genes	73.7

Bold identifies the highest descriptive gene-panel average, including ties at displayed precision; paired contrast intervals above are not ranked. The 2k-HVG and 8,192-HVG labels are requested caps of up to 2,000 and 8,192 support-selected genes, with lower realized widths possible (Methods). These descriptive twenty-draw averages differ from the three-draw headline benchmark in Table 1. Random-panel controls assess the delivered panels and cannot isolate every effect of feature selection or gene coverage.

Table S4: Operator ablation and the prespecified decision.

Panel	Support/class	Full minus no operator (pp; 95% CI)
PBMC coarse	1	+0.20 [-0.23, 0.64]
PBMC coarse	10	-0.50 [-1.24, 0.24]
PBMC coarse	100	+2.69 [2.35, 3.02]
PBMC fine	1	+0.61 [0.19, 1.03]
PBMC fine	10	+4.18 [3.48, 4.87]
PBMC fine	100	+6.81 [6.36, 7.26]
ZEBRA development	1	-0.29 [-0.60, 0.01]
ZEBRA development	10	+1.97 [0.99, 2.95]
ZEBRA development	100	+2.07 [1.77, 2.37]
ZEBRA held-study	1	-1.35 [-1.98, -0.73]
ZEBRA held-study	10	+2.71 [1.39, 4.04]
ZEBRA held-study	100	+2.33 [1.86, 2.80]

Twenty paired support draws, one training seed per arm, all genes. Intervals are exported paired 95% t intervals over support draws. Full and no-operator checkpoints share the parent, prior and task stream. These later descriptive evaluations do not replace the prespecified decision below.

Operator structure across support budgets.

Panel	Support/ class	None	Diagonal	Full	Full minus diagonal (pp; 95% CI)
PBMC coarse	1	59.81	59.09	60.02	+0.92 [0.39, 1.45]
PBMC coarse	10	85.84	85.69	85.34	-0.34 [-0.86, 0.18]
PBMC coarse	100	89.14	88.89	91.83	+2.94 [2.64, 3.24]
PBMC fine	1	51.30	51.38	51.91	+0.53 [0.13, 0.93]
PBMC fine	10	75.64	78.48	79.81	+1.33 [0.87, 1.79]
PBMC fine	100	82.22	85.66	89.03	+3.37 [3.06, 3.68]
ZEBRA development	1	46.62	46.45	46.33	-0.12 [-0.49, 0.24]
ZEBRA development	10	61.36	63.20	63.33	+0.13 [-0.48, 0.75]
ZEBRA development	100	63.70	64.74	65.77	+1.03 [0.89, 1.18]
ZEBRA held-study	1	66.53	65.84	65.17	-0.67 [-1.08, -0.25]
ZEBRA held-study	10	88.59	91.11	91.31	+0.19 [-0.32, 0.70]
ZEBRA held-study	100	92.10	93.57	94.43	+0.86 [0.40, 1.32]

Values are mean balanced accuracy (%). The diagonal ablation is a trained per-gene-variance-only operator arm; it shares the parent, prior and task stream with the full-operator arm. At one support per class, the inference operator gate is zero: differences between these separately trained arms reflect their learned states, not an active inference-time operator. The printed matrices use all genes. The source-value index retains additional HVG and random-panel analyses.

Prespecified three-draw readout.

Panel	Support/class	Full minus none (pp)	Harm threshold (pp)
PBMC coarse	10	-1.56	1.50
PBMC coarse	100	+3.02	1.50
ZEBRA held-study	10	+3.30	2.72

The overall test was not supported because coarse-PBMC ten-support accuracy crossed the prespecified harm threshold (N3). Turning the full operator off at inference isolated a harmful -0.94-point operator contribution in that setting. Its donor-enumeration interval has approximately 80% coverage, not 95%, and is not used as a 95% interval here. Three frozen support draws and four query donors informed the PBMC test; the later twenty-draw estimates describe support sensitivity rather than overturning the decision.

Table S5: Complete ten-candidate prior screen.

Arm/base	Change	r42	r43	r44	Mean	Decision
LR / b0	Task-mixture restriction	+1.63	+1.62	+2.61	+1.95	Keep
A5 / lr	Fewer/weaker markers and closer sibling classes	-1.52	-1.05	-1.54	-1.37	Drop: guard
A6 / b0	Remove signed-covariance task lane	-0.19	+0.74	+1.16	+0.57	Neutral
A7 / b0	Donor dominance	+0.16	-0.21	+0.65	+0.20	Unresolved
A1 / b0	More donor-rich tasks	-0.0008	-0.0080	-0.0133	-0.0074	Routing only
A3 / a1	Sex mechanisms	-0.0089	-0.0089	+0.0357	+0.0060	Routing only
A4 / lr	Weak metadata tasks	+0.0110	-0.0347	-0.0325	-0.0188	Routing only
A2 / a1age	Adult-age prior	+0.0083	+0.0000	+0.0000	+0.0028	Routing only
A8a / b0	Study-wide gene absence	+0.52	+0.54	+1.64	+0.90	Drop: guard
A8d / b0	Study-specific detection	-0.43	+0.60	+0.87	+0.35	Not testable

All changes are improvement-signed relative to the indicated base. LR, A5, A6, A7, A8a and A8d use balanced-accuracy percentage points; A1, A3 and A2 use donor AUC differences on a 0–1 scale; A4 uses negative log-loss differences (base minus arm, nats). A7 uses the held-study panel; A8a/A8d use low-support cortex; LR/A5/A6 use all low-support core panels. A1/A3/A2 respectively target donor, worse-sex and adult-age AUC. The Kumo-small proxy uses 10,000 cellular tasks, 625 updates and 2k HVGs. The three columns are task-stream replicates from one model seed, not independent model-seed replications. A5 and A4 are compared with LR; A3 with A1; A2 with the matched age-task stream. Other candidates use the pipeline control b0.

LR was the only retained change: its keep threshold was 1.63 points and effective noise band 1.09 points. A5 failed the low-support guard. A7 remained unresolved; omitting the prescribed two extension replicates departed from the frozen rule, although the required keep count was no longer attainable. A1–A4 routing labels do not establish successful donor-level or probability-prediction improvements: those endpoints were outside the proxy's trusted decision scope. A8a failed the log-loss guard (0.099-nat deterioration against a 0.03-nat limit). A8d was not testable because measured headroom (1.257 points) was below the 1.5-point keep threshold; its two extension replicates were omitted under the documented shortcut because the required keep count was no longer attainable. A8d replicate r43 skipped one invalid task, an authorized deviation made after the other two replicate readouts. None of these routing, unresolved or headroom-limited outcomes is evidence that its biological mechanism does or does not work.

Stored-zero mechanism readout (percentage points).

Arm	Mean gap shrinkage	Noise band	Proxy headroom	Keep threshold
A8a	0.072	0.620	1.257	1.500
A8d	0.071	0.620	1.257	1.500

Gap shrinkage is the reduction in the accuracy difference between removing unmeasured genes and retaining stored zeros. The gap did not shrink beyond the noise band; the reference row repeats the three-replicate aggregate, not a single-replicate estimate.

Table S6: Restricted task mixture versus standard CellPFN continuation.

Panel	Support/class	Accuracy change (pp; 95% CI)	Log-loss change (nats; 95% CI)
PBMC coarse	1	+1.11 [0.45, 1.77]	-0.332 [-0.400, -0.265]
PBMC coarse	10	-1.47 [-2.12, -0.81]	+0.416 [0.337, 0.496]
PBMC coarse	100	-1.56 [-1.94, -1.19]	+0.374 [0.357, 0.391]
PBMC fine	1	+2.42 [1.61, 3.23]	-0.742 [-0.773, -0.710]
PBMC fine	10	+1.06 [0.26, 1.86]	-0.159 [-0.172, -0.145]
PBMC fine	100	+0.50 [0.22, 0.78]	-0.047 [-0.051, -0.042]
ZEBRA development	1	+2.56 [1.11, 4.02]	-0.054 [-0.205, 0.097]
ZEBRA development	10	+1.34 [0.35, 2.33]	+0.281 [0.198, 0.363]
ZEBRA development	100	+0.33 [0.19, 0.48]	+0.629 [0.606, 0.652]
ZEBRA held-study	1	+3.49 [1.73, 5.26]	-0.251 [-0.363, -0.139]
ZEBRA held-study	10	+1.38 [0.72, 2.05]	+0.249 [0.186, 0.312]
ZEBRA held-study	100	+0.24 [-0.02, 0.49]	+0.346 [0.304, 0.388]

Restricted task mixture (GL1) minus standard (GS1) at 100,000 updates; twenty paired support draws; all genes. Positive accuracy change is favourable, whereas positive log-loss change is unfavourable. Intervals are exported paired 95% t intervals over support draws. Both arms continue the same checkpoint for 4,000 updates with one model seed. Log loss measures probabilistic prediction and is not exclusively a calibration measure. The standard task-family mixture is 80% metadata, 15% cell type and 5% donor identity. The restricted mixture samples cell-type and donor-identity tasks with nominal probabilities 75% and 25%, omits metadata tasks and skips signed-covariance tasks. This is a joint change in task families, mechanisms and training exposure, not a test of metadata-task learnability.

Realized training exposure during continuation.

Task stream	Mean cells per update
Restricted task mixture	2667.8
Standard	1596.6

Task stream	Metadata tasks	Cell-type tasks	Donor-identity tasks
Standard	12816	2398	786
Restricted task mixture	Excluded	12001	3999

The restricted task-mixture continuation processed approximately 1.67 times as many cells per update. Its gains therefore cannot be attributed to task filtering alone under matched cell exposure. Higher-support log loss worsened on three of four panels but improved on fine PBMC.

Table S7: Simulator realism diagnostics against public biological references.

Statistic	Prior	Synthetic median [5th, 95th percentile]	Real median [minimum, maximum]
Strongest marker (log2 FC)	Baseline prior	1.936 [1.408, 3.497]	7.100 [3.310, 9.762]
Strongest marker (log2 FC)	Cellular prior	6.232 [4.273, 10.225]	7.100 [3.310, 9.762]
Class separation (chance-adjusted BA)	Baseline prior	0.022 [0.001, 0.120]	0.775 [0.525, 0.916]
Class separation (chance-adjusted BA)	Cellular prior	0.881 [0.522, 0.994]	0.775 [0.525, 0.916]
Within-study donor SD (log2)	Baseline prior	0.331 [0.213, 0.489]	0.290 [0.197, 0.391]
Within-study donor SD (log2)	Cellular prior	0.404 [0.291, 0.528]	0.290 [0.197, 0.391]

Synthetic quantile ranges and observed real ranges are descriptive distributions, not confidence intervals. The reference contains public PBMC, cortex and kidney datasets; ROSMAP is excluded. Class separation uses nearest-class-mean balanced accuracy at ten supports per class with eight classes in both synthetic and primary real references.

Sampling units and synthetic tail frequencies by prior.

Statistic	Prior	Synthetic banks	Synthetic units	Real units	Synthetic above real max (%)
Strongest marker (log2 FC)	Baseline prior	72	2016	142	0.0
Strongest marker (log2 FC)	Cellular prior	96	2688	142	8.3
Class separation (chance-adjusted BA)	Baseline prior	72	72	8	0.0
Class separation (chance-adjusted BA)	Cellular prior	96	96	8	37.5
Within-study donor SD (log2)	Baseline prior	96	768	11	37.9
Within-study donor SD (log2)	Cellular prior	128	1024	11	58.9

Units differ by statistic: class-pair comparisons for marker effects, dataset-level summaries for separation and donor-effect units for within-study SD. Units from one simulated bank are not independent biological replicates. The real reference helped calibrate the prior, so this is a reference-comparison diagnostic rather than independent validation. The cellular prior supplied only the final 16.7% of scheduled CellPFN training episodes. Improved moment agreement is not itself evidence of a causal performance benefit.

Table S8: DS-LOO fidelity and support sensitivity.

Case	DS-LOO full: Spearman	DS-LOO none: Spearman	IG64 full: Spearman	DS-LOO full: relative error (%)
PBMC, 10 supports, draw 0	0.997	0.997	0.789	2.91
PBMC, 10 supports, draw 1	0.998	0.997	0.802	2.31
PBMC, 100 supports	0.998	0.995	0.794	1.56
Cortex development, 10 supports	0.993	0.988	0.809	0.48
Kidney, counts	0.986	0.979	0.867	4.88
Kidney, processed	0.988	0.989	0.937	1.17

Corrected-margin DS-LOO and 64-step integrated gradients (IG64) are compared with the same exact query-only single-gene deletion margin target and evaluated gene set in each case, in fp32, using correctly classified cells. IG64 is available for the full-operator model in all six cases. Spearman values summarize within-cell rank correlations, aggregated within classes and then across classes. Relative L_2 errors are computed per class and summarized by the median across classes. These are six case summaries per model, not minima across every cell. No one-shot or held-study cortex attribution validation is included. A positive deletion-margin attribution means that the observed gene favours the predicted class over its fixed runner-up; a negative value opposes that margin and does not mean biological downregulation. Setting an already zero query value to zero cannot measure information carried by its absence.

Comparison	Range of class-mean top-20 overlap
Different PBMC support choices	0.431–0.563
Full versus no operator	0.663–0.878

Overlap is the fraction of common genes among the twenty highest-ranked genes, averaged over classes for each comparison. The support comparisons include both a changed support draw at ten cells per class and a changed support budget; they are not a pure seed-only sensitivity analysis. Agreement with model knockout measures explanation fidelity, not causal biological validation.

Three highest-ranked genes for each explained cell class.

Panel	Cell class	Rank 1	Rank 2	Rank 3
PBMC	B	CD79A	MS4A1	BANK1
PBMC	CD4 T	MAL	IL7R	TRAC
PBMC	CD8 T	CD8B	CD8A	TRBC2
PBMC	DC	FCER1A	CLEC10A	HLA-DQA1
PBMC	Mono	SERPINA1	FTL	CD14
PBMC	NK	KLRF1	GNLY	IL2RB
PBMC	other	GNG11	PF4	PPBP
PBMC	other T	GZMK	TRGC2	KLRB1
ZEBRA	Glia	NPAS3	PCDH9	PPP2R2B
ZEBRA	Immune cell	MBNL1	FMN1	DPYD
ZEBRA	Neural	FGF12	RGS7	RBFOX1
ZEBRA	Vascular cell	PTPRG	MALAT1	LAMA2

Genes are selected directly by their class-mean corrected DS-LOO margin rank, independently of the curated marker checks below. PBMC uses 100 supports per class, draw 0; ZEBRA development uses ten supports per class, draw 0. These are model-ranked features conditional on the support set and class-discrimination target; they are not all canonical lineage markers and do not establish causal effects.

Known-marker ranks in the PBMC reference case.

Cell class	Marker	Corrected DS-LOO rank
B	CD79A	1
B	MS4A1	2
CD4 T	IL7R	2
CD8 T	CD8A	2
DC	FCER1A	1
DC	CLEC10A	2
Mono	CD14	3
NK	GNLY	2
NK	NKG7	7
other	PPBP	3
other T	CD3E	5
DC	LILRA4	4392
Mono	LYZ	33354

The reference uses 100 supports per class and support draw 0, explaining 191 correctly classified query cells across eight broad classes and 33,538 genes. Ranks refer to class-mean corrected margin attribution; smaller is higher. Familiar lineage markers[40] rank highly in several classes, but LYZ in monocytes and LILRA4 in pooled dendritic cells illustrate that marker identity does not guarantee a high rank for this class-discrimination target. These are selected descriptive marker checks, not an enrichment test or causal validation. Gene identities were assigned after inference.

For example, monocyte LYZ ranks 33354 for corrected-margin DS-LOO but 5 for predicted-class-logit DS-LOO. The estimand and attribution variant therefore matter when interpreting marker ranks.

Contextual marker ranks in ZEBRA cell-type classification.

Cell class	Marker	Corrected DS-LOO rank
Glia	PLP1	8
Neural	SYT1	7
Vascular cell	FLT1	5
Vascular cell	CLDN5	16
Immune cell	PTPRC	20710

ZEBRA development, ten supports per class, draw 0. PLP1, SYT1 and CLDN5 provide contextual cell-identity checks[41]. The immune-class ranking uses 24 correctly classified cells; the source report gives low immune-class recall (approximately 0.17), which was not reanalysed here. The ranking therefore characterizes this selected subset. These are cell-type attribution ranks; they are not explanations of the sex-classification experiments in Table S9.

Table S9: Exploratory sensitivity analyses and provisional application.

Panel	Support/class	Processed cortex BA (%)	Reconstructed-count BA (%)
ZEBRA development	1	41.4	51.4
ZEBRA development	10	63.6	63.4
ZEBRA development	100	66.1	64.9
ZEBRA held-study	1	62.8	65.9
ZEBRA held-study	10	91.9	92.9
ZEBRA held-study	100	94.7	93.8

Mean CellPFN balanced accuracy over three support draws. Processed and reconstructed-count cortex have different gene sets (21,158 versus 12,325), and some support identities change. These comparisons therefore do not isolate normalization, count reconstruction or stored-zero effects; the processed representation remains the primary panel.

Removing unmeasured genes while retaining processed values.

Panel	Support/class	Accuracy change (pp; 95% CI)
ZEBRA development	1	+3.62 [0.42, 6.83]
ZEBRA development	10	+0.64 [0.38, 0.90]
ZEBRA development	100	+0.14 [-0.10, 0.38]
ZEBRA held-study	1	+1.04 [-0.67, 2.75]
ZEBRA held-study	10	+0.51 [0.03, 0.99]
ZEBRA held-study	100	-0.47 [-1.66, 0.71]

These Stage-0 contrasts remove genes not measured in every study without changing the retained expression values or re-normalizing their denominator. Values and paired 95% t intervals are the exported summaries over ten support draws. This is a separate diagnostic from the three-draw count-reconstruction comparison above.

Post hoc classifier sensitivity. Standardizing scGPT embeddings before logistic regression changed the PBMC mean from 69.5% to 73.5%. This diagnostic is not substituted for the declared baseline in Table 1.

Post hoc hard-tier boundary. Task-mixture restriction in proxy training changed held-study cortex neuronal-subtype accuracy at ten supports by -5.83 percentage points across three task streams (-6.62 to -4.56). Hard-tier tasks were defined after the initial screen and do not change its frozen labels. A8d's hard-tier readout contains only two task-stream replicates.

Early task-mixture gene-view sensitivity at update 98,304.

Gene view	Support draws	PBMC low-support change (pp)
All genes	3	+0.344
8,192 HVGs	10	-0.892

Changes are GL1 minus GS1, equally weighting coarse and fine PBMC at one and ten supports per class. Different draw counts and the early checkpoint prevent treating this as a final, isolated explanation of the task-mixture tradeoff.

Provisional ROSMAP application: sex-matched microglial CERAD discrimination gave donor AUC 0.745 [0.700, 0.790]. The supplied interval summarizes ten repeated splits with the export's correction; it is not independent cohort replication. Label corrections remain pending, so this is not a final or confirmatory biological result.

PBMC annotation levels and fine-grained subtype tasks.

Task	Classes	1 support/class	10 supports/class	100 supports/class
Broad PBMC types (L1)	8	62.3	84.2	91.9
Fine PBMC types (L2)	20	53.9	81.8	90.5
CD4 T siblings (L2)	4	36.3	69.2	85.6
T-cell siblings (L2)	9	27.4	64.4	83.8
CD4 T siblings (L3)	9	23.4	56.3	79.6
CD8 T siblings (L3)	7	39.5	69.4	84.4
B-cell siblings (L3)	6	31.6	63.6	79.8
NK-cell siblings (L3)	6	36.2	64.3	73.0
T-cell siblings (L3)	17	25.0	55.7	76.8
All finest PBMC types (L3)	35	39.8	68.4	78.4
Mean of eight subtype tasks	–	32.4	63.9	80.2

Values are mean balanced accuracy (%) for the reference CellPFN checkpoint with a requested cap of up to 8,192 support-selected HVGs; actual widths can be lower. Broad/fine context rows use the E-fast ten-, ten- and five-draw readouts at 1, 10 and 100 supports, respectively, rather than the three-draw main-table means. The subtype aggregate uses the corresponding ten/ten/five case design. L2 and L3 denote the supplied second and third annotation levels; sibling tasks restrict classification to a related group. These eight tasks were defined after the initial prior-screen readouts and are exploratory; they are not a formally defined "easy versus difficult" partition. The broad/coarse and fine primary panels remain in Table 1. The available checkpoint summaries do not provide CellPFN support-draw uncertainty, so no interval is inferred from the proxy-model case records. Different label sets and class counts prevent treating cross-task accuracy differences as isolated changes in difficulty.

Finer ZEBRA annotation tasks at the reference checkpoint.

Task	Classes	1 support/class	10 supports/class	100 supports/class
Finer cortex types, development	7	67.8	76.3	79.3
Neural siblings, held studies	3	52.5	62.1	69.6

The development task distinguishes Astrocyte, Endothelial, Excitatory neuron, Inhibitory neuron, Microglia, OPC and Oligodendrocyte. The held-study neural task distinguishes Excitatory neuron, Inhibitory neuron and Pvalb neuron. These post hoc exploratory tasks use the same reference checkpoint and requested cap of up to 8,192 support-selected HVGs, with ten, ten and five support draws at 1, 10 and 100 supports per class. Values are supplied checkpoint mean balanced accuracies (%); no confidence intervals are available or inferred. They are separate from the eight-task PBMC subtype aggregate and the four-group core cortex panels, and do not constitute independent cohorts.

Original ZEBRA sex-classification case.

Scoring unit	Balanced accuracy (%)	Log loss (nats)
Query cells	100.0	0.409
Aggregated GSM samples	100.0	0.398

This single Sayed-study oligodendrocyte case used 200 support cells from 12 GSM samples and 50 query cells from ten disjoint GSM samples, with 21,158 processed genes. GSM separation does not guarantee donor separation. One support draw was evaluated; the perfect observed accuracy does not imply an uncertainty-free population estimate. This case is separate from the four core cell-type panels and is excluded from their averages.

Later unit-level sex evaluation: positive control and broader task.

Support units/class	XY-eligible task AUC	Broader sex task AUC
4	0.977 [0.872, 1.000]	0.507 [0.391, 0.617]
8	0.993 [0.949, 1.000]	0.544 [0.453, 0.617]
16	1.000 [1.000, 1.000]	0.538 [0.441, 0.633]
32	1.000 [1.000, 1.000]	0.539 [0.426, 0.656]

Values are mean [minimum, maximum] unit-level AUC over ten support splits at the same reference CellPFN checkpoint, using a requested cap of up to 8,192 support-selected HVGs, ten support cells per unit and twenty query cells per unit. Actual gene widths can be lower. Units are persons where resolved; Sayed age-by-sex GSM groups remain a proxy, and cross-study person overlap is unknown. The XY-eligible positive control requires measured XIST and at least four Y-linked genes; this eligibility rule does not identify genes used by the model. The broader task remains close to chance. Splits reuse persons and are not independent cohort replications, so observed ranges are shown without confidence intervals. Support budgets of 16 and 32 units per class exceed the training support range. These later tasks are distinct from the original cell-level case above.

Sex-specific gene attribution remains unresolved. XIST and Y-linked genes define eligibility; they are not measured attribution hits.[O06]

Supplementary Methods

Simulator parameter distributions and reusable software interfaces

Supplementary Table S10 describes the principal cellular-prior configuration. Intervals denote distributions across independently seeded experiments, not confidence intervals. The baseline prior and named ablation arms retain separate configuration records; values in this table must not be retroactively assigned to every training phase. Integer counts are drawn on inclusive integer ranges. Class-signal, residual-cell, group-nuisance, class/domain-interaction and trajectory energies are log-uniform, as are intermittent-marker log-effect/amplitude SD and the dispersion jitter's log-SD. Fractions, correlation magnitude, hierarchy/subtype parameters, structured-state parameters and donor-library hyperparameters use uniform draws on their stated ranges. Program activation probabilities are uniform draws per program. Program size is lognormal around the drawn median with log-SD 0.7, clipped to 2–$G/4$ genes; positive state-program loadings have lognormal log-SD 0.5. Nonlinear-metadata density and energy are uniform, whereas kernel-reuse size is an integer draw. The resolved configuration and source-to-equation audit accompany the manuscript.

Table S10: Principal cellular-prior parameters.

Component	Specification	Interpretation
Experiment topology	4 studies; 1 batch/study; 16 generated donors; training bank uses 8 donors	Group identities are generated before episode selection.
Training bank	$8 \times 8 \times 256$ cells; 8,192 genes	Donor × cell type × cells per type; 16,384 cells/bank.
Biological families	Mean programs : marker multimodality : signed covariance, 2:1:1	Fixed family schedule for consecutive bank roots.
Baseline body	Lognormal log-SD 1.5–2.3	Procedural anonymous gene abundance.
High-abundance component	Probability 0.001–0.025; log-location 4.0–6.3; log-scale 0.3–1.0	Maximum normalized baseline gene share 0.1.
Mean programs	24 programs; gene fraction 0.015–0.16; at least 24 genes; loading 0.30–3.00	Class energy 0.03–2.0.
Marker-family stable programs	24 programs; gene fraction 0.03–0.08; at least 24 genes; loading 0.5–2.0	Stable class energy 0.05–2.0.
Intermittent markers	Gene fraction 0.002–0.05; at least 8 genes; activation 0.15–0.85	Log-effect 0.35–2.0; log-amplitude SD 0.05–0.60.
Class hierarchy	2–4 levels; level SD ratio 0.3–1.0	Marker fraction 0.003–0.02; log-effect 1.75–3.25.
Class bounds	Metric proportional to baseline$^{0.5}$; saturation 4.5	Maximum class-template gene share 0.35; 16 admission attempts.
Hidden subtypes	Activation probability 0.3–0.8; 2–4 subtypes; energy ratio 0.2–2.0	Dirichlet-weighted centring after saturation.
Covariance family	6 programs; gene fraction 0.05–0.08; loading 0.9–1.1	Energy 0.07–0.12; pair correlation magnitude 0.75–0.95.
Cell-state energy	0.03–0.12 for each family	Baseline-weighted energy metric.
Structured cell state	Fraction 0.7–0.95; 8–20 programs; median size 80–250 genes	Activation 0.1–0.5; shared fraction 0.7–0.95.
Structured-state modifiers	Quality weight 0.02–0.1; abundance power 0.6–1.0	Mean compensation 0.3–0.7; donor program-logit SD 0.5–1.0.
Group nuisance energy	Study 0.003–0.03; batch 0.003–0.03; donor 0.002–0.015	Gene-wise shifts independent of target labels.
Class/domain interactions	Enabled probability 0.25	Study energy 0.001–0.035; donor energy 0.001–0.045.
Trajectory	Probability 0.25; energy 0.001–0.040	Position-jitter SD 0–0.20.

Table S10, Principal cellular-prior parameters (continued)

Component	Specification	Interpretation
Linear metadata	Selected-target multiplier 0.25; other-metadata multiplier 0.6	Scaling follows effect normalization; donor aging is separate.
Nonlinear metadata	Gene density 0.02–0.05; shared-kernel group size 20–100; energy 0.003–0.02	Direct age kernels excluded while the clock is active.
Age and sex	Age 60–90, rounded; binary sex threshold 0.5 and steepness 10	Public variables derived from generated copula metadata.
Aging coordinate	Fixed depth 1,500; normalized log1p counts; 105 maximum steps	Ages mapped using fixed bounds 5–110.
Aging noise	Initial SD 0.01; step SD 0.005; donor-rate log-SD 0.2	Prepared donor-specific clipped maps; pre-normalization fold bounds e^{-1}–e^{1}.
Aging gene-rate mixture	Noisy fraction 0.1–0.3; noisy log-SD 0.5–1.0	Quiet scale maximum 0.2; donor jitter SD 0.1.
Age-dependent cell noise	Gene fraction 0.05–0.2; increasing fraction 0.6–0.67	Susceptible-class fraction 0.5–0.8; slope 0.05–0.25; expected log-state shift preserved.
Directional age effects	Gene fraction 0.02–0.1; negative fraction 0.6–0.9	Energy 0.001–0.01.
Library hierarchy	Global log-mean 7.53; base study/batch/cell SD 0.443/0.45/0.50	Study/batch multipliers 0.3–1.0; cell multiplier 1.0–1.65.
Donor library	SD 0.2–0.45; global log-offset −0.6–0	Minimum library 50.
Mean-dependent dispersion	$\phi_0 = 0.1$, $m_0 = 0.1$, exponent 1.2; amplitude log-uniform 0.05–2.0	Class-template reference; gamma-shape bounds 0.01–160; lognormal jitter SD 0.3–0.8.
Observation dropout	Disabled in principal configuration	Study-specific detection and absence belong to named ablations.
Bank composition	Global Dirichlet concentration 1–3; donor concentration 2–10	At least 64 available rows; age-composition class fraction 0.2–0.3 and log-effect −0.5–0.5.
Low-count bank QC	Redraw cells below 20 observed counts, up to 8 technical redraws	Keep the highest-count candidate; banks containing any zero-count cell are rejected.

Reusable components and execution boundaries. The package separates biological generation, observation policies, delivery and logging. The provider protocol comprises `cell_types`, `gene_ids` and `expression_for_donor(meta_values)`; a configured dotted Python factory can supply an external implementation. Generic effects implement `forward(x)` on cell-by-gene tensors and can be composed at plate, batch and study levels. Their order is scientifically meaningful: a terminal count draw cannot be interchanged with a continuous intensity transformation. Optional capabilities in Supplementary Table S11 are software interfaces or separately evaluated interventions, not additional unreported components of the principal prior.

Table S11: Simulator software capabilities and their relationship to the evaluated system.

Component	Implemented capability	Usage boundary
Experiment assembly	Configurable cell hierarchy, donor metadata, studies, batches and plates	Principal tasks use the controlled four-study topology.
Biological providers	Procedural task families and external donor-expression factories	CellPFN uses the mixed cellular task provider; external providers are an extension interface.

Table S11, Simulator software capabilities (continued)

Component	Implemented capability	Usage boundary
Observation effects	Library scaling, overdispersion, dropout, ambient mixture, doublets, panel masks, $5'/3'$ bias and FFPE decay	General simulator options; the principal task-owned observation model is specified in Methods.
Controlled prior interventions	Additional donor cohorts, donor-private on-switches, study detection and gene absence	Used only where the named ablation activates the corresponding block.
Spatial/noise utilities	Three-dimensional value, fractal, Gaussian-feature, Perlin and simplex noise fields	Standalone utilities; no spatial training claim is made for CellPFN.
Calibration modules	Transcriptome, library, dispersion, ambient, dropout, doublet and study-effect fitting; diagnostic interfaces	Availability does not imply all fitted effects were active in training.
PyTorch delivery	Dense cell tensors through `IterableDataset`; count-contract validation	CPU generation; local DataLoader workers disabled.
Independent producers	Spawned CPU processes, bounded prefetch, deterministic task order and timeouts	Implemented by the CellPFN adapter around independently initialized DGPs.
Task reuse	Immutable count banks, metadata and content receipts	Different targets and support/query views reuse fixed biological counts.
Native acceleration	Optional C++ batched latent draws with exact reference checks and fallback	Specific shared-topic backend, not the complete cellular-prior kernel.
Annotated export	AnnData dense or sparse observed counts; H5AD/Zarr/Loom loggers	Expected profiles and intensities remain dense in the sparse export.
Ground truth and provenance	Separate public/hidden loggers, runtime seeds, configuration and array hashes	Hidden state is excluded from model-facing episode payloads.
Serialization	Reconstruct governed task datasets from configuration and seed	Active task datasets cannot be safely cloned through pickling across processes.

A minimal stream and export interface. The following example uses the implemented CellPFN adapter signatures. It takes an experiment configuration supplied by the caller, fixes its seed, obtains 64 cells and writes an annotated count matrix. The full calibrated training configuration additionally fixes task-family parameters and target construction. This is a usage example, not a throughput measurement.

```
from pathlib import Path
import anndata as ad
import pandas as pd
from hashprompt.dataset.dataset import (
    Config, build_cell_stream, synchronize_task_root_seed)

def simulate_chunk(config_path, output_path, seed=42):
    cfg = Config(Path(config_path))
    cfg = synchronize_task_root_seed(cfg, seed)
    stream = build_cell_stream(
        cfg, micro_batch_size=64, device="cpu", reseed=False)
    batch = next(iter(stream))
    counts = batch["observed_counts"].numpy()
    out = ad.AnnData(X=counts, obs=pd.DataFrame(batch["meta"]))
    out.layers["expected_profile"] = (
        batch["expected_profile"].numpy())
    out.layers["expected_counts"] = (
        batch["expected_counts"].numpy())
    out.write_h5ad(output_path)
    return out
```

For a governed task configuration, the explicit synchronization call binds the task root and top-level seed. Production exporters additionally preserve gene annotations, public/hidden information boundaries and count/runtime receipts. Generation remains on CPU; downstream model code can transfer its selected tensors to an accelerator. The example requires the simulator, CellPFN adapter and AnnData dependencies.

Memory and numerical representation. Let B be the simulation microbatch size, C the number of cells retained in an output chunk and G the gene count. The active generator holds several dense $B \times G$ arrays plus experiment-level class/program/donor state; the constant depends on which biological blocks and hidden artifacts are retained. Direct streaming avoids retaining all previously generated cells. Dense collation allocates three four-byte $C \times G$ arrays, requiring $12CG$ bytes before metadata; creating the returned AnnData copies these arrays, giving approximately $24CG$ bytes while both buffers and output are alive, before other temporary allocations. The sparse collator reduces storage of observed counts but still retains two dense float32 arrays, requiring $8CG$ bytes in addition to sparse counts, indices and temporary copies. These are storage calculations, not measured peak-RAM values.

The current chunk exporter aggregates 100 microbatches of 10,000 cells. Its million-cell output size is an implementation setting; runtime and feasibility depend on gene count and output layers. The CellPFN fixed-bank path uses smaller materialization batches of 64 cells and releases simulator objects after bank production. One 16,384-cell, 8,192-gene bank occupies 512 MiB as int32 or 256 MiB when losslessly stored as uint16, excluding metadata; active and prefetched banks add to this memory. Prepared donor-aging maps cost $O(TG)$ once per donor for T walk steps and $O(G)$ per subsequent application, preserving each intermediate clipping operation. No claim of GPU simulation, universal differentiability or end-to-end sparse sampling follows from the PyTorch interface.

CellPFN feature definitions and tensor dimensions

The feature definitions below specify the count and processed branches of the same predictor. Let $b_{ig} = \mathbf{1}(x_{ig} > 0)$, $f_i = G^{-1} \sum_g b_{ig}$, and let $\bar{b}_g$ and $\bar{b}_{cg}$ denote support-wide and class-specific detection fractions. Support-wide and class-specific population moments of ℓ are $(\bar{\ell}_g, s_g)$ and $(\bar{\ell}_{cg}, s_{cg})$, respectively. Define $\tilde{s}_g = \max(s_g, 10^{-6})$. This floored support-wide standard deviation is used both in equation 1 and in the neural input features. Class-specific standard deviations are descriptive features and can be zero. For count inputs, define $v_{ig} = (\ell_{ig}/10, z_{ig}/4, b_{ig}, \sqrt{Gp_{ig} + 10^{-6}})$. The processed view omits the fourth component. All row depths and detection fractions use the delivered gene panel, rather than a gene chunk.

The support-gene input consists, in order, of v_{ig}, $(\bar{\ell}_g/10, \tilde{s}_g/10, \bar{b}_g)$ and $(\bar{\ell}_{y_i g}/10, s_{y_i g}/10, \bar{b}_{y_i g})$. This gives ten count channels or nine processed channels. The support summary is $\tilde{v}_{ckg} = \sum_{i:y_i=c} a_{cki} v_{ig}$. The

relation input begins with $[v_{qg}, \tilde{v}_{ckg}, v_{qg} \odot \tilde{v}_{ckg}, |v_{qg} - \tilde{v}_{ckg}|]$. It then appends the three support-wide and three class-specific gene statistics above. In count mode, its row-level features are $\log(1 + d_q)/12$, f_q, the summary-weighted support $\log(1 + d_i)/12$, and the summary-weighted support f_i; processed mode contains only the two detection features. Both modes finish with $\log(1 + n_c)/\log(101)$ and the alignment channel $\frac{1}{2}(v_{qg,1}\tilde{v}_{ckg,1} + v_{qg,2}\tilde{v}_{ckg,2})$. Thus the dimensions are $16 + 6 + 4 + 1 + 1 = 28$ and $12 + 6 + 2 + 1 + 1 = 22$.

Support-row and relation reductions use population RMS, not standard deviation, and log-mean-exp, not unnormalized log-sum-exp. Gene chunks accumulate sums, sums of squares, maxima and log-sum-exp before normalization by the total gene count. The reference configuration uses 1,024 genes and 32 query–class pairs per chunk. The padded summary dimension is eight, of which at most four entries per class are active. Relation representations therefore have logical dimensions $Q \times C \times 8 \times G \times 224$, but are reduced in chunks rather than materialized in full. Summary tokens have dimensions $Q \times C \times 8 \times 448$, program tokens $Q \times C \times 8 \times 448$, and logits $Q \times C$. Masks exclude inactive summaries from cross-attention. No positional encoding is attached to gene columns, support rows or query rows. Joint permutation of support and query genes leaves the prediction unchanged up to floating-point arithmetic; support-row permutations and relabelling classes have the corresponding invariant/equivariant action.

The gene–gene fit uses the class means of clipped standardized expression, which generally differ from standardizing the un-clipped class means. For a support-variable gene with zero within-class variance, the shared diagonal floor remains active. A gene constant across every support row is excluded even if a query value differs. Eigenvectors are obtained from the $N \times N$ support Gram matrix and mapped to gene coordinates; the implementation does not construct a dense $G \times G$ covariance. Float64 factors are detached from automatic differentiation, while the gene channel and three pair scalars are returned in float32. The operator's two learned integration maps remain differentiable. The operator therefore represents statistical dependence estimated from the prompt, without interpreting covariance as a directed or causal gene network.

Table S12: CellPFN architecture and learned parameter counts. The diagonal and full gene–gene arms have the same learned parameter shapes and counts. The covariance estimator itself has no learned parameters.

Component	Dimensions and computation	Parameters
Support-gene encoder	Count $10 \to 64$; processed $9 \to 64$; GELU; shared $64 \to 64$; layer normalization	5,632
Support-row projection	Four 64-channel reductions; $256 \to 448$; GELU; layer normalization	116,032
Support-row residual blocks	Three blocks: layer normalization; $448 \to 1,792 \to 448$; GELU	4,826,304
Summary query projection	$448 \to 448$	201,152
Summary key projection	$448 \to 448$	201,152
Summary queries	Eight learned vectors of width 448	3,584
Summary roles	Eight learned vectors of width 448	3,584
Relation encoder	Count $28 \to 448$; processed $22 \to 448$; GELU; shared $448 \to 224$; layer normalization; includes 448 operator-stem parameters	124,768
Relation reduction projection	Four 224-channel reductions; $896 \to 448$; GELU; layer normalization	402,752
Program queries	Eight learned vectors of width 448	3,584
Program cross-attention	Width 448; eight heads	804,608
Cross-attention normalization	Layer normalization, width 448	896
Observation-mode embeddings	Two learned vectors of width 448	896
Program transformer	Two pre-normalized layers; width 448; eight heads; feed-forward width 1,792; GELU; no dropout	4,828,544
Direct classifier	Program mean/max; $896 \to 448 \to 1$; GELU	402,305
Operator scalar head	Three scalars; $3 \to 448 \to 896$; GELU	404,096

Supplementary Table S12 (continued)

Component	Dimensions and computation	Parameters
Total, diagonal or full operator	Includes the operator's 448-weight gene stem and 404,096-parameter scalar head	**12,329,889**
Total, no operator	Same backbone with both operator integration paths absent	**11,925,345**

Training distributions, objectives and checkpoint reconstruction

For zero-based bank index j, define $p_j = \mathrm{clip}((j-1,000)/9,000, 0, 1)$. The probabilities of binary metadata, categorical metadata, cell type and donor identity are $(1-0.4p_j, 0.2p_j, 0.15p_j, 0.05p_j)$. Binary and categorical tasks have two and four classes, respectively; cell-type and donor-identity tasks draw uniformly from two, four, six and eight classes. Requested panel width is 2,048 before bank 1,000 and thereafter has probabilities $(1-0.7p_j, 0.35p_j, 0.35p_j)$ for 2,048, 4,096 and 8,192 genes. These are requested widths; the memory cap can reduce them. Natural-frequency query allocation imposes a floor of ten cells per class, caps allocations by available cells and adjusts rounding to retain the nominal $100C$ total when feasible. Metadata targets select the bank's dominant non-age covariate with probability one-half and otherwise draw an eligible metadata field. Age can be a target from bank 10,000; its generative effects need not be absent before it becomes a supervised target.

In the baseline-prior phase, admission is applied with probability $1 - j/10,000$. On separate diagnostic cells (16 support and 16 query cells per class), the better of the empirical-Bayes negative-binomial and combined raw/standardized-expression similarity controls must achieve accuracy at least $1/C + 0.10$. A nearest-class-mean rule using only support-standardized log depth and detection fraction must score strictly more than 0.025 lower than that best control. Balanced diagnostic class sizes make accuracy equal balanced accuracy here. A rejected episode has zero training weight. The filter is inactive from bank 10,000. It does not use CellPFN predictions or the cells used to calculate its training loss. Admission labels indicate performance of these diagnostic rules, not a proof that rejected tasks contain no biological information.

Let $\mathcal{A}_t$ be accepted tasks in optimizer opportunity t, and Q_e the query count in task e. The objective is

$$L_t = \frac{1}{|\mathcal{A}_t|} \sum_{e \in \mathcal{A}_t} \left[-\frac{1}{Q_e} \sum_{q=1}^{Q_e} \log p_\theta(y_{eq} \mid X_{eS}, y_{eS}, x_{eq}) \right]. \tag{18}$$

The normalization is over tasks, rather than pooling all query cells across tasks; within a task it is ordinary query cross-entropy, without class weighting. No auxiliary distillation, reconstruction or contrastive objective is applied to the direct model. For $t \le 8,000$, the learning-rate multiplier is $t/8,000$; subsequently it is $0.1 + 0.45[1 + \cos(\pi(t-8,000)/192,000)]$. The continuation retains this global schedule rather than restarting the warm-up. Its operator group multiplies the backbone rate by $10 \min(1, \max(0, t - 80,000)/1,000)$.

The baseline-prior stage includes the specified retention of the early biological task distribution in one-half of banks after bank 5,000. This is a sampling component of that stage. In the cellular-prior continuation, the early-prior flag is retained in task records but does not activate the baseline expression generator. Each continuation restores model weights, Adam moments, random-number states, task positions and receipt chains. This state restoration matters because parameters can have different optimizer step counters after admission filtering or mode-specific updates. The full and diagonal operators retain the parent tensor values and add only their declared, initially silent integration paths. Full training reproducibility therefore requires the staged recipe and parent state, rather than training a randomly initialized final architecture for 384,000 episodes on a single stationary prior.

The task-mixture comparison adds 16,000 episodes (4,000 update opportunities) to the same full-operator endpoint. Its targeted distribution assigns cell type and donor identity in a 3:1 ratio and skips signed-covariance banks. The reference continuation keeps the parent distribution. Both retain the architecture and optimization policy. The targeted arm keeps 500 banks but skips 166 intervening candidates, so its bank index reaches 12,665; its training receipt's 12,500 cumulative accepted-bank count must not be interpreted as the largest candidate index. The arms also deliver different cell counts per update, which is part of this

distributional intervention.

Table S13: Training and adaptation recipes. Counts distinguish independent simulated banks, scheduled tasks, tasks contributing loss and optimizer opportunities. The nominal 5,000-task adaptation milestone is update 313 and therefore contains 5,008 cellular task slots.

Setting	Evaluated configuration
Simulated bank	Eight donors × eight cell types × 256 cells, 8,192 genes; first 32 cells per donor–type reserved for diagnostics
Reuse and batch	32 task views per bank; four interleaved banks; four scheduled tasks per optimizer opportunity
Support sizes	$B = 1, 10, 100$: probability 0.25 each; $B = 2, 4, 8, 16$: 0.04 each; $B = 32, 64$: 0.045 each
Query and element budget	Nominal 100 queries per class; frequency-weighted allocation for cellular-prior cell-type tasks; at most 12.5 million delivered matrix elements
Observation mixture	70% raw-count and 30% processed-expression tasks
Initial baseline-prior stage	10,000 banks; 320,000 scheduled and 212,828 loss-bearing episodes; 80,000 optimizer opportunities
Cellular-prior continuation	2,000 additional banks; 64,000 scheduled and loss-bearing episodes; 16,000 additional optimizer opportunities
Principal endpoint	12,000 banks; 384,000 scheduled and 276,828 loss-bearing episodes; 96,000 optimizer opportunities
CellPFN random seeds	Model seed 42; task-stream seed 15113; one model initialization lineage
CellPFN optimizer	AdamW; $\beta = (0.9, 0.999)$; weight decay 10^{-4}; groupwise gradient norm clipping at 5
CellPFN learning rate	Base 10^{-4} through opportunity 40,000, 10^{-5} thereafter; 8,000-step warm-up; cosine floor 0.1; horizon 200,000; operator group 10× backbone rate with a 1,000-step ramp from 80,000
CellPFN numerical policy	Float32 parameters/statistics/losses; bfloat16 neural autocast; float64 support covariance; gene/pair chunks 1,024/32
Task-mixture continuation	Additional 500 accepted banks and 16,000 episodes; 4,000 opportunities; standard distribution versus cell type:donor identity $3:1$ with signed-covariance banks excluded
Tabular initialization	Released TabICL, 27,552,258 parameters; Kumo small, 27,458,266; Kumo large, 213,668,250
Tabular task stream	Shared 40,000 cellular tasks plus 10,000 generic replay tables; 16+4 tasks per update; 2,500 updates
Tabular task geometry	Support 1/10/100 in exact thirds; 32 queries per class; at most eight cellular classes; support-fitted 2k HVGs or at most 100 principal components
Tabular optimization	AdamW; model seed 42; $\beta = (0.9, 0.999)$; zero weight decay; peak rate 3×10^{-5}; 125 warm-up updates; cosine to zero at 2,500; norm clipping 1; float32 training
Adaptation milestones	Updates 313, 625, 1,250 and 2,500; approximately 5,000 and exactly 10,000, 20,000 and 40,000 cellular task slots
Prior-content proxy endpoint	Update 625 (10,000 cellular tasks), retaining the 2,500-update learning-rate horizon; task-draw replicates distinguished from model-seed replicates

Tabular preprocessing and adaptation interfaces

Cellular episodes are converted to tables with support rows first, followed by query rows. The view builder normalizes counts to log1p counts per 10,000 before selecting genes using support-derived mean-binned dispersion statistics. The PCA view standardizes support rows, clips standardized values to $[-10, 10]$ and retains $\min(100, N-1, H)$ components, where H is the selected feature count. Each component is oriented so its largest absolute loading is positive. Queries undergo the corresponding support-fitted transformations. For TabICL, preprocessing returns an array with dimensions $E \times (N+Q) \times H$, support labels $E \times N$, and query targets $E \times Q$, where $E \in \{1, 2\}$ is the ensemble-member count and H is the resulting feature width. Feature permutations, shifted class-label patterns and normalization settings are recorded, allowing probabilities to be returned to the original class order. These arrays are produced deterministically from the task plan, independently of worker completion order.

Kumo consumes the same support-fitted cellular views but applies its own support-fitted default recipe, rather than TabICL's transforms. Its four recipe members and the one/two differentiated members are distinct quantities. One forward/backward pass is accumulated per selected member; their losses are averaged within task before the task average. Gradient checkpointing is applied to model blocks. Deployment evaluation uses the released library route, including the class-count wrapper, with eight or 16 members for small or large models. Thus a single training member, a cellular episode, an optimizer update and an inference ensemble are not interchangeable exposure units. The adaptation interfaces retain checkpoint identity, task and replay array hashes, the consumed task chain, preprocessing settings and invalid-task reasons. External model libraries supply the pretrained model implementations; hashprompt supplies reproducible cellular task construction, model-specific training adapters and evaluation wrappers.

Software reference and experimental provenance

The software description is bound to hashprompt main commit `c06f99d` and pyScIntelligentDesign main commit `6c4d2d0`. The principal exported experiment is bound to model-source commit `9e50f8e` and simulator commit `76ba3a3`. Optional capabilities available in the software reference are identified separately from enabled experimental components. Experiment records retain resolved configuration, model/checkpoint identifiers, initialization and task seeds, bank and task receipts, preprocessing settings, and source-array hashes. Full source identifiers and the claim-to-implementation crosswalk accompany the reproducibility record. These bindings distinguish the software resource from the exact configuration evaluated in each result.

Evaluation settings and statistical definitions

Table S14 records the evaluated comparator contracts. The primary benchmark's gene views are not interchangeable: a full-gene CellPFN result, a support-selected tabular result and a gene-vocabulary embedding result compare practical inference routes rather than holding every input coordinate fixed. Hyperparameters below are the supplied evaluation settings, not proposed retuning.

Table S14: Recorded comparator inputs and fitting settings.

Comparator	Input and fitting data	Recorded settings
TabICLv2	Support-selected genes; labelled context	`tabicl` 2.2.0; released weights and package-default inference for the primary gene view. Support-only PCA is a separate sensitivity view.
TabPFN-3.5	Support-selected genes; labelled context	`tabpfn` 9.0.0; released weights and package defaults, including the feature guard. Non-commercial research use.

Table S14 continued

Comparator	Input and fitting data	Recorded settings
Kumo Tabular	Support-selected genes; labelled context	Official 1.0.0 weights; 16 ensemble members for large, eight for small. Default recipe, at most 500 columns per member, logit averaging; float16 CUDA autocast with float32 weights.
Geneformer-V2-316M	Rank-value tokens mapped by Ensembl identifier; frozen mean embeddings	At most 4,094 genes per cell. Support-standardized L2 logistic regression: $C = 1$, LBFGS, maximum 2,000 iterations. Geneformer 0.1.0 and Transformers 4.46.0.
scGPT whole-human	Gene-symbol vocabulary; frozen unit-length CLS embeddings	At most 1,199 nonzero genes per cell, seeded subsampling above this limit, 51 value bins. Unscaled L2 logistic regression: $C = 1$, LBFGS, maximum 10,000 iterations. scGPT 0.2.4.
Expression logistic regression	Support-selected expression genes or PCA; support fit only	Support standardization, $C = 1$, LBFGS, maximum 2,000 iterations, tolerance 10^{-6}, random state 0.
CellTypist	Full processed gene panel; support fit only	CellTypist 1.7.1; $C = 1$, LBFGS, maximum 1,000 iterations, seed 42; no additional feature selection, stochastic-gradient fitting or query majority voting. Native one-versus-rest probabilities normalized for NLL/ECE without changing the best-match class.
SingleR	Full log-normalized panel; support cells as reference	BiocPy SingleR 0.5.0; classic pairwise markers, Spearman scoring, quantile 0.8, fine-tuning threshold 0.05. R SingleR 2.12.0 provided a separate concordance check.
scANVI	Up to 2,000 support-selected genes and counts	scvi-tools 1.4.2; scVI 400 epochs then scANVI ten epochs, training fraction 0.9, batch size 128, no class-balanced replacement sampling (`n_samples_per_label=None`), seed 0. Separate inductive support-only and transductive support-plus-unlabelled-query conditions.
LDA diagnostics	Processed full-gene or 2,000-gene inputs; support fit	Diagonal, Ledoit–Wolf shrinkage and rank-32 low-rank covariance variants; recorded comparisons at ten and 100 supports. Full-gene low-rank comparisons use the Woodbury implementation.

Frozen inputs and feature provenance. The ZEBRA subsample SHA256 is `428fe0a064327cb1d513367e5bee2ab08a1cf0a5ba93d40e9b2197299550ebd9`; the case builder checks this identity and the frozen evaluation-manifest identity before reusing its splits. The PBMC preparation joins exact source barcodes and retains the donor time codes without interpreting them as independently sampled subjects. Antibody quality-control totals in GEO metadata differed from the reproduction snapshot even though the audited antibody values and labels agreed; no protein-derived quantity entered inference. Feature selection is conditional on the labelled support matrix, except for the pre-existing kidney feature panels and the assay-availability mask explicitly described above.

For cortex count reconstruction, let $z_g = \exp(x_g) - 1$ for a processed expression row and let u be its smallest positive entry. Candidate counts were z_g/u. A row was rejected if any coordinate was more than 0.001 from an integer or if its reconstructed CP10k/log-transform round trip differed by more than 2×10^{-6}. Reconstruction preceded masking of unavailable genes. The 1,322 Yang Cortex cells failing this rule were excluded from eligible supports; existing support rankings were filtered, preserved in order and extended deterministically from the eligible original support pool. The original query cohorts were retained exactly. These checks support a conditional integer-lattice reconstruction; they do not convert inferred counts into

independently observed raw reads.

Sex-task units and label resolution. The study-matched oligodendrocyte positive control required XIST and at least four of RPS4Y1, DDX3Y, UTY, KDM5D and EIF1AY to have been measured. Its 46 female–male unit pairs spanned eight studies; fourteen pairs entered each query split. The broader task added four Lim pairs and used sixteen query pairs. Common measured-gene universes were 16,708 and 12,668 genes, respectively, before support-selected 8,192-gene evaluation. Both tasks used the same four nested support-unit budgets, but 16 and 32 units supplied 160 and 320 cells per class, beyond the training support range. The joint change in studies and gene coverage is not a controlled removal of sex-linked genes. Means and ranges over reused-person splits are descriptive.

PBMC fine-task eligibility used the original donor quotas. Candidate tasks were retained when

$$\frac{R/100 - 1/K}{1 - 1/K} \geq 0.20 \quad \text{and} \quad R < 85,$$

where K is the class count and R is the larger mean balanced accuracy of nearest-class-mean and logistic-reference classifiers at 100 supports. These references used support-standardized 2,000-gene expression. The rule did not consult CellPFN/proxy outputs, but it did consult reference predictive performance and query labels. It therefore does not create an independent external test set. Nested labels and overlapping cells also prevent interpreting an eight-task mean as eight biological replicates.

Metric definitions. With K query classes, class sizes n_c and predictions $\hat{y}_i$, balanced accuracy was

$$\mathrm{BA} = \frac{1}{K} \sum_{c=1}^{K} \frac{1}{n_c} \sum_{i:y_i=c} \mathbf{1}(\hat{y}_i = c).$$

Every declared class had to occur among the queries; the scorer rejects silent class dropping. Probabilities were obtained from class logits, and NLL was $-n^{-1} \sum_i \log \max(p_{i,y_i}, 10^{-12})$. ECE was $\sum_{j=1}^{15} (n_j/n) |\mathrm{accuracy}_j - \mathrm{confidence}_j|$, using confidence equal to the maximum class probability and bins $[(j-1)/15, j/15)$ for $j = 1, \ldots, 15$ with the final right edge included. Empty bins contributed zero. Reported BA is multiplied by 100; BA differences are percentage points. AUC remains on the zero-to-one scale.

For a paired diagnostic contrast over D support draws, d_s was computed within draw before averaging. The conditional 95% interval was $\bar{d} \pm t_{0.975,D-1} s_d/\sqrt{D}$. This construction was used only where the analysis supplied a ten- or twenty-draw interval; it was not applied to the three-draw headline benchmark or reused-person sex controls. For the provisional ROSMAP analysis, the reported standard error used the Nadeau–Bengio correction[55], $s\sqrt{1/10 + n_{\mathrm{query}}/n_{\mathrm{support}}}$, where s is the standard deviation over splits; the interval multiplier was 2.262. Its units are donors under the stated split design, not cells. The available aggregate export does not permit independent reconstruction of the underlying donor assignments or label corrections.

Conditional pool and operator updates for DS-LOO. For each query and its fixed predicted/runner-up classes, concatenate the pooled relation representations into P and the additive operator pair embeddings into h, with fixed-pair margin $H(P, h)$. Holding query depth and detected fraction fixed, replace gene g's relation vector r_g by its zero-expression counterpart r_g^0. If G genes are supplied, the changes in the componentwise mean and RMS are

$$\Delta P_g^{\mathrm{mean}} = \frac{r_g^0 - r_g}{G}, \qquad \Delta P_g^{\mathrm{RMS}} = \sqrt{\max\left(\frac{\sum_j r_j^2 + (r_g^0)^2 - r_g^2}{G} + \epsilon, 0 \right)} - \sqrt{\frac{\sum_j r_j^2}{G} + \epsilon}.$$

Top-two maxima and their indices give the exact maximum after replacement. Writing $L = \log \sum_j \exp(r_j)$ componentwise, the normalized log-sum-exp change is

$$\Delta P_g^{\mathrm{LSE}} = \log \max\left[1 - \exp(r_g - L) + \exp(r_g^0 - L), 10^{-30}\right].$$

These replacements are exact conditional on the other relations and row-level scalars being fixed.

For the full operator, let Σ denote the fitted covariance in equation 11, restricted to support-variable coordinates, and let v be the support-fitted inverse diagonal scale and

$$A = I - U \operatorname{diag}\left[1 - (1+\lambda)^{-1/2}\right] U^{\mathsf{T}}, \qquad w_c = A\{v \odot (z - \mu_c)\}.$$

At fixed depth, setting gene g to zero changes its standardized coordinate by $\delta_g = z_g^0 - z_g$. DS-LOO updates the own-gene whitening coordinate as

$$w_{cg}^{(g)} = w_{cg} + A_{gg} v_g \delta_g,$$

then applies the original clipping to $[-8, 8]$, scaling by $1/4$ and support gate. Changes $A_{jg} v_g \delta_g$ in other coordinates $j \neq g$ are omitted. Thus the low-rank operator introduces an explicit approximation beyond head linearization.

The operator's scalar branch is updated separately. For every class c, its squared Mahalanobis distance changes according to

$$M_c^{(g)} = \max\left\{M_c + 2\delta_g [\Sigma^{-1}(z - \mu_c)]_g + \delta_g^2 [\Sigma^{-1}]_{gg}, 0\right\}.$$

The implementation recomputes the original across-class centering, normalization, asinh transformation, gate-weighted recentering and gated scalar features from these distances, then reevaluates the nonlinear pair multilayer perceptron to obtain Δh_g. The resulting uncorrected approximation is

$$\widehat{I}_g^{\text{conditional}} = -\nabla_P H(P,h)^{\mathsf{T}} \Delta P_g - \nabla_h H(P,h)^{\mathsf{T}} \Delta h_g.$$

The second term is absent in the no-operator model. Pair-embedding reevaluation is exact for the conditional scalar inputs; the final scorer remains linearized. Selected-gene exact head replay evaluates $H(P+\Delta P_g, h+\Delta h_g)$ and therefore isolates head linearization while retaining the conditional-state approximations.

First-order depth and detection correction. Let $d = \max(\sum_j x_j, 1)$ in count mode and $q = G^{-1} \sum_j \mathbf{1}(x_j > 0)$. With expression held fixed, differentiate the reconstructed fixed-pair margin with respect to independent depth and detection leaves at the original query. The implemented corrected estimate is

$$\widehat{I}_g^{\text{corrected}} = \widehat{I}_g^{\text{conditional}} + \mathbf{1}_{\text{count}} \frac{\partial m}{\partial d} x_g + \frac{\partial m}{\partial q} \frac{\mathbf{1}(x_g > 0)}{G}.$$

The depth derivative passes through the differentiable full query-side operator, including depth-induced changes to its whitening coordinates. It does not restore fixed-depth off-diagonal deletion effects. The correction is first order and does not eliminate nonlinear remainders or effects of clipping and the depth floor. Processed-mode inputs receive only the detected-fraction correction. Operator algebra uses float64, relation pools and head evaluation use float32, and TF32 is disabled.

The recovered historical consistency checks matched the reconstructed pooled representation and base operator channels exactly for all twelve case–model combinations. The largest recorded final-logit discrepancy was 1.04×10^{-3} for the reference model and 1.53×10^{-4} for the no-operator model. These checks are recorded execution diagnostics; they do not make the finite-deletion approximation exact.

Validation genes were drawn from unions of each method's class-level top twenty, the bottom twenty for integrated gradients and the operator readout, and twenty random expressed genes, augmented with previously untested DS-LOO top/bottom genes. DS-LOO scored every gene, while exact validation covered this selected union. The analysis compared gradient-times-input and integrated-gradients variants[39] with the same finite-deletion target; numerical agreement is evidence of fidelity to that target, not a proof that the attribution methods estimate identical scientific quantities. For the principal corrected-DS-LOO comparison, Spearman correlations were computed across the original exact-deletion gene set for each cell, summarized by the median within each class and then the median over classes. Constant or undefined correlations were excluded by the recorded finite-value rule. Relative Euclidean error was computed within each class after

flattening the gene-by-selected-cell arrays, $\|\widehat{I} - I\|_F / \|I\|_F$, and summarized by the median across classes. Additional DS-LOO-selected extreme genes augmented separately labelled uncorrected validation statistics; they did not augment the principal corrected statistic. Exact head replay and finite query deletion were kept distinct. Deletion-curve reversal denotes the original predicted-versus-runner-up margin becoming negative, not a recomputed global-argmax label change. Validation quantifies agreement with the specified finite-deletion target and supplies no analytical error bound or causal biological interpretation. The six cases include two representations of the same kidney cells and therefore are not six independent cohorts. Cell-level deletion curves and support-plus-query deletion experiments were kept separate because the latter refit the support statistics.

Supplementary Figures

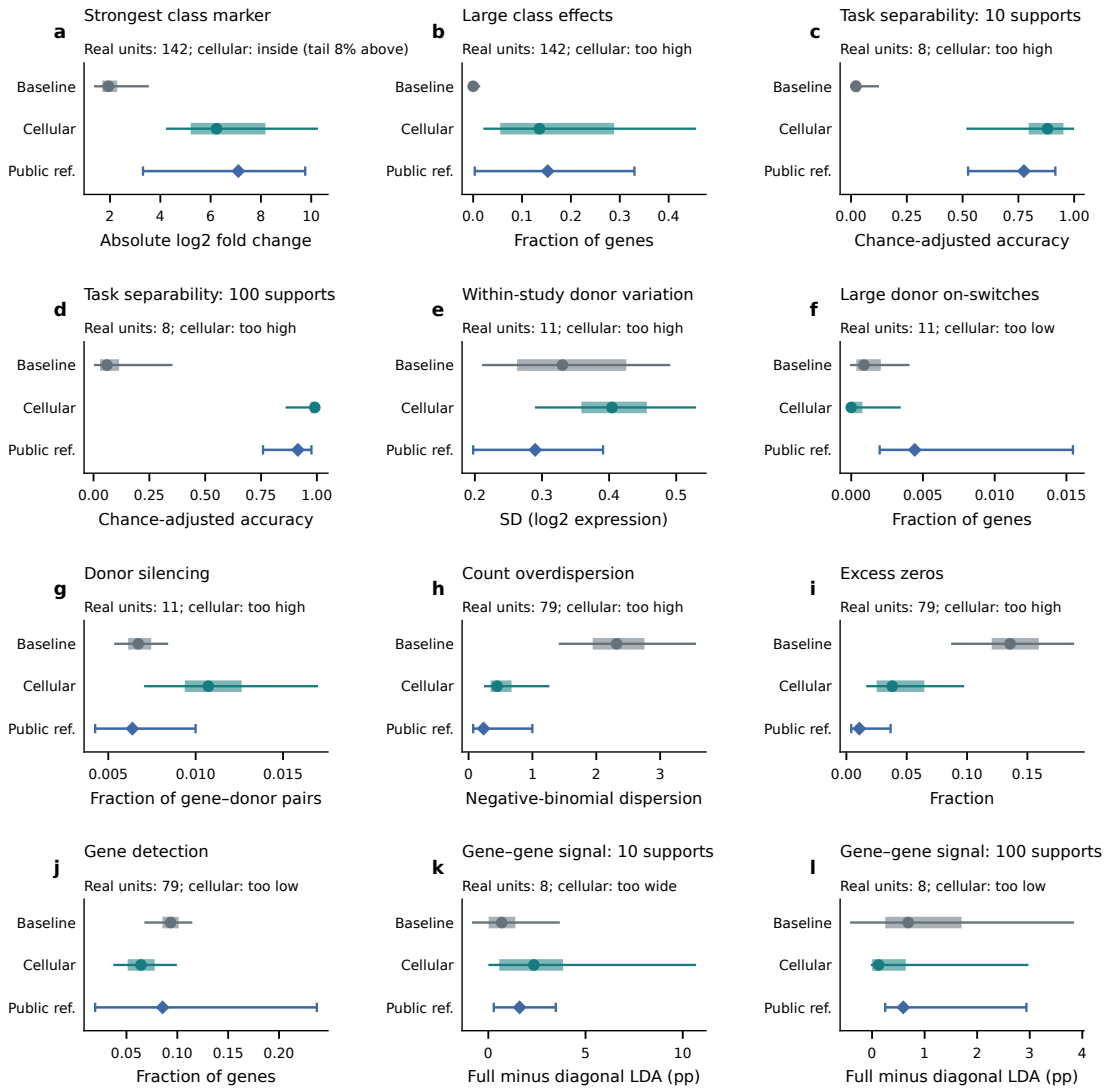


Figure S1: Calibration covers biological effects while retaining distributional mismatches. a,b, Strongest class marker and fraction of differentially expressed genes with absolute $\log_2$ fold change above two. **c,d**, Chance-adjusted nearest-class-mean balanced accuracy on eight-class tasks with ten or 100 supports per class. **e–g**, Within-study donor variation, tenfold on-switches and silencing. **h–j**, Count dispersion, excess zeros and detection. **k,l**, Full minus diagonal LDA accuracy; pp, percentage points. Synthetic circles, thick segments and whiskers show median, interquartile range and fifth–95th percentiles. Public-reference diamonds and whiskers show median and range; none are confidence intervals. Real units number 142 (a,b), eight (c,d,k,l), eleven (e–g) and 79 (h–j). The cellular prior uses 128 fresh banks; class-effect/separability summaries exclude signed-covariance banks, leaving 96. Source Data give exact counts. Real units summarize matched draws whereas synthetic units are individual draws, preventing matched-tail inference. Public references are PBMC, human cortex and kidney; ROSMAP is excluded. These datasets informed calibration and evaluation, so this is not independent real-data validation. Coverage labels retain the exported judgments. CellPFN received this prior in only its final 64,000 of 384,000 scheduled episodes.

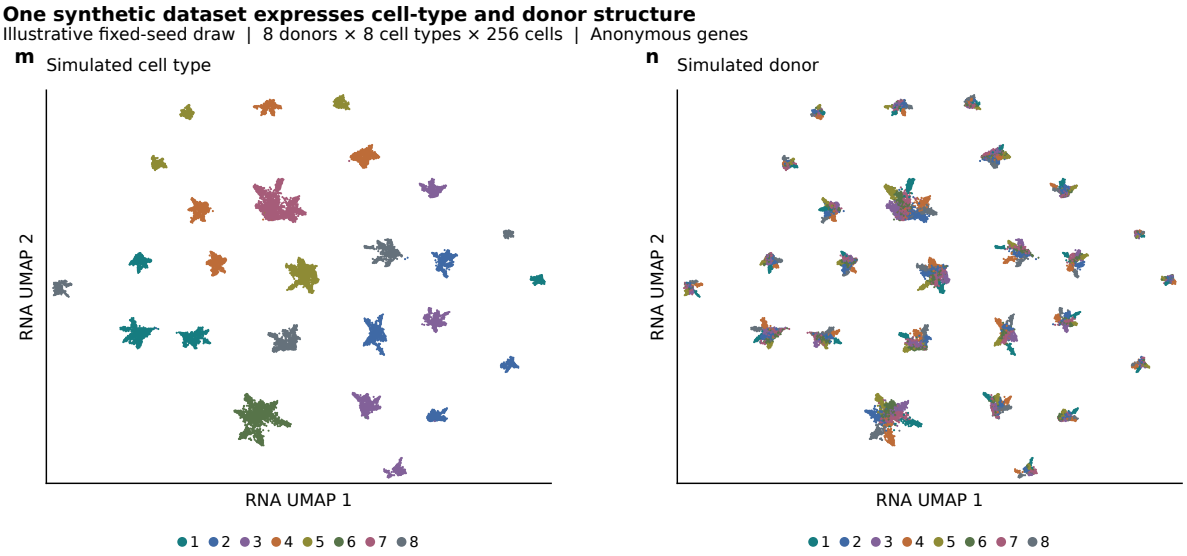


Figure S1 (continued). Illustrative RNA geometry of one predefined cellular-prior draw. m,n, Identical unsupervised UMAP coordinates coloured by simulated cell type or donor. All 16,384 cells comprise eight donors, eight types and 256 cells per combination, with 8,192 anonymous genes. The mean-program bank (root 44; seed 42) was selected before generation. Log1p counts-per-10,000 and 2k HVGs supplied 50 principal components, 15 Euclidean neighbours and UMAP (minimum distance 0.3; seed 42). Labels did not determine geometry. This single draw is illustrative, not a realism assessment. Source Data provide coordinates, selected genes and the configuration receipt.

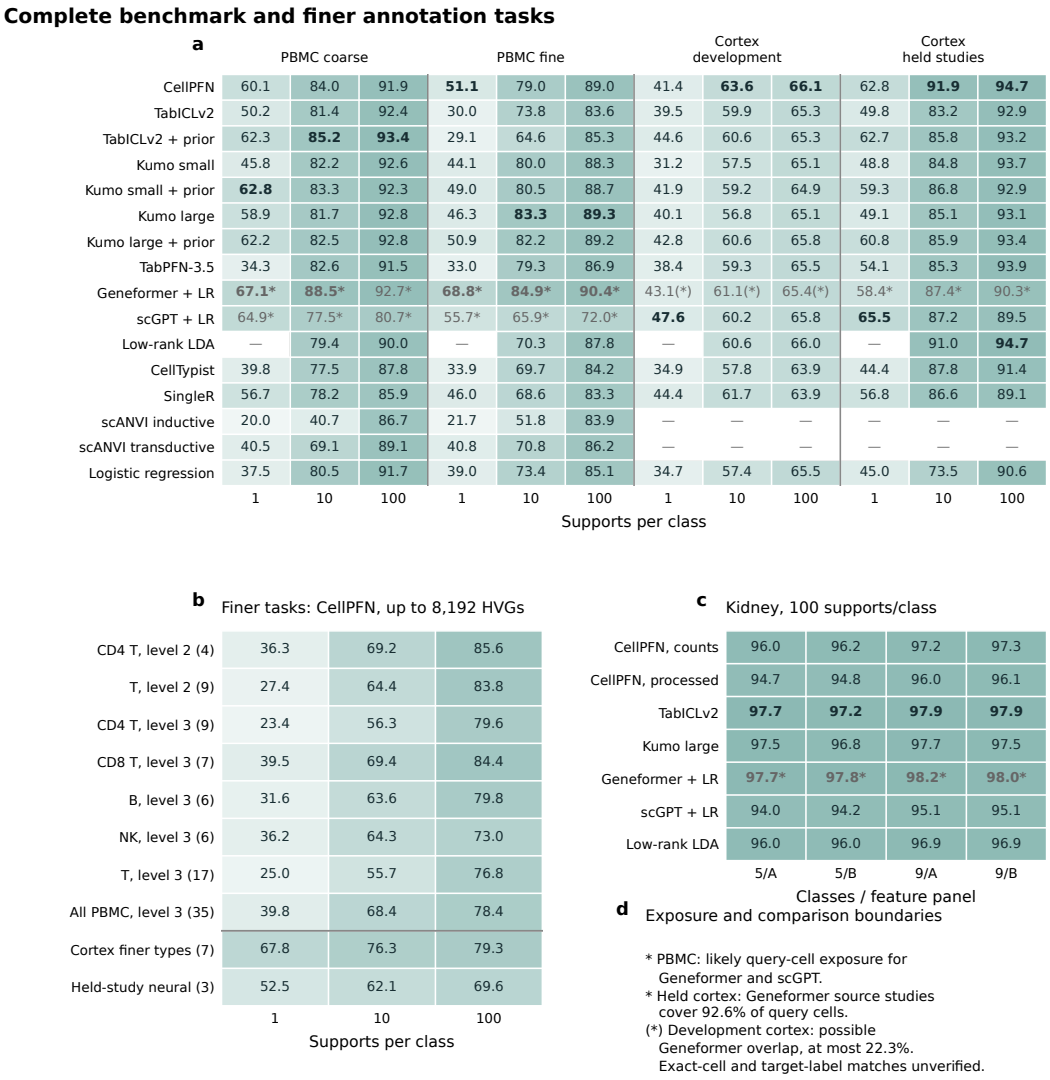


Figure S2: Complete biological benchmark and finer annotation tasks. a, Mean balanced accuracy (%) over three support draws. Bold identifies the best overall and best unflagged means within each column, including displayed-precision ties. Gray stars follow Table 1; unflagged entries do not certify no exposure. Dashes denote unavailable evaluations. Input routes and constrained scANVI training follow Methods; transductive scANVI uses query expression. TabPFN-3.5 is used for non-commercial research only. **b,** CellPFN on eight PBMC subtype and two finer cortex tasks with up to 8,192 HVGs. Parentheses give class counts. Supplied means use ten, ten and five support draws at 1, 10 and 100 supports/class; unavailable draw-level values preclude inferred intervals. These post hoc tasks reuse existing collections and differ in class sets, rather than forming a controlled difficulty scale. Cortex development has seven types; held-study cortex has three neural siblings. **c,** Kidney cell-type accuracy across donor-held-out folds, separately for four 4,096-gene panels (five/nine classes; feature sets A/B). Both CellPFN input modes and Geneformer exposure flags are retained. **d,** Likely PBMC cell exposure versus cortical source overlap. Bounds of 92.6% and at most 22.3% concern overlapping/potentially overlapping query sources; exact-cell and label matching remain unverified. Complete fold scores accompany the figure.

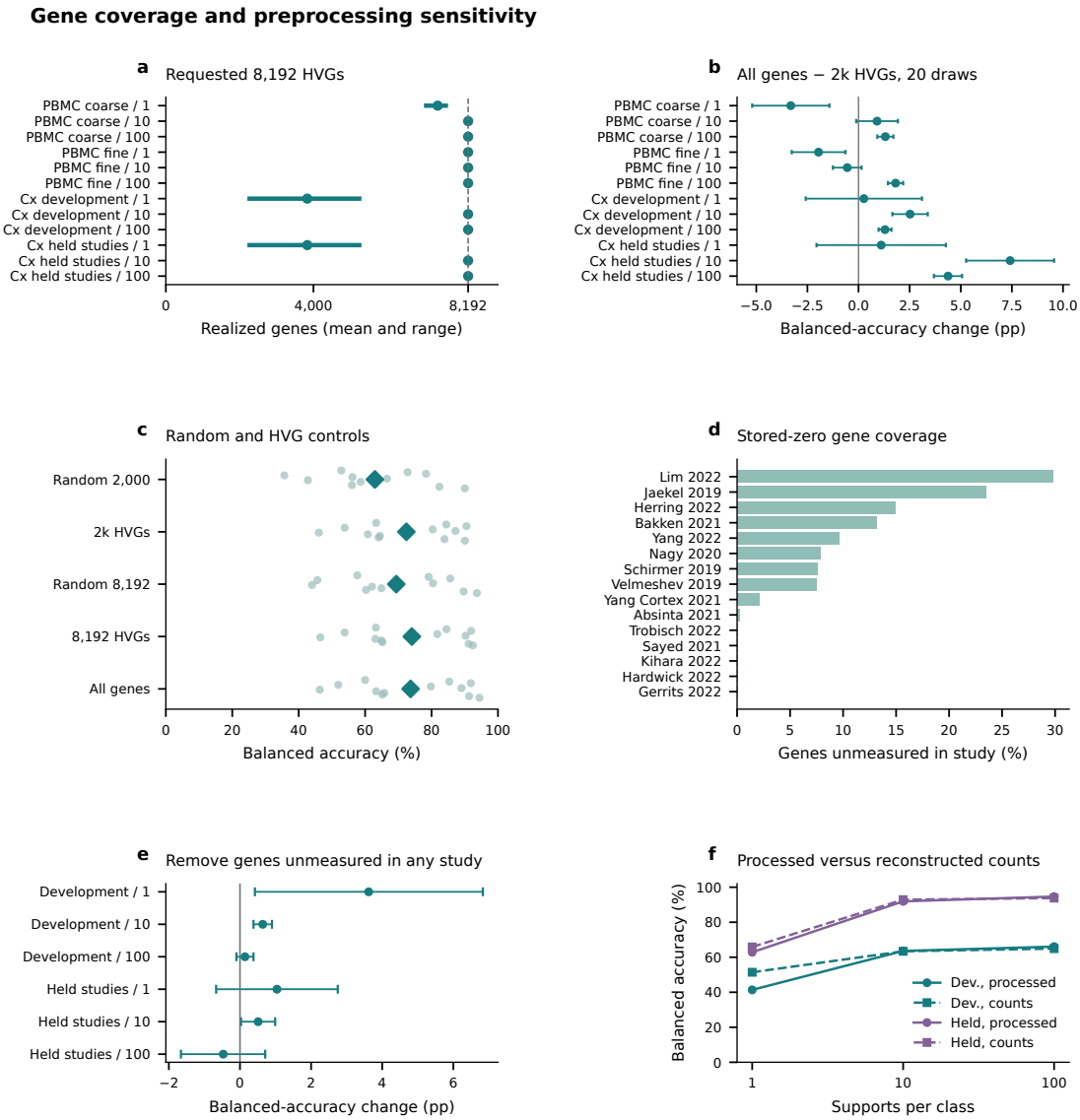


Figure S3: Gene coverage and preprocessing sensitivity. All panels use the reference CellPFN checkpoint. **a,** Realized genes under the 8,192-HVG cap: means and ranges over three support draws. Labels give panel/supports per class; fewer eligible genes can occur at one-shot support. Source data retain 2k-HVG and all-gene widths. **b,** All-gene minus 2k-HVG balanced-accuracy changes, with exported paired means and 95% t intervals over twenty support draws. Intervals condition on the checkpoint and queries, excluding training and donor-population uncertainty. **c,** Random-gene and HVG controls. Small points are twenty-draw means for twelve panel–budget conditions; diamonds equally average conditions. Their spread describes task heterogeneity, not an inferential interval. HVG labels denote caps. **d,** Fraction of 21,158 merged cortex genes unmeasured per study and stored as zero. Source data identify support, query and separate sex-case study roles. **e,** Changes after removing genes unmeasured in any study, preserving retained processed values and their normalization denominator: paired means and 95% t intervals over ten draws. **f,** Three-draw means for processed cortex versus reconstructed counts. Gene coverage (21,158 versus 12,325) and some support identities differ, preventing an isolated normalization, reconstruction or stored-zero comparison. "Dev." and "Held" denote development and held-study cortex; processed cortex remains primary.

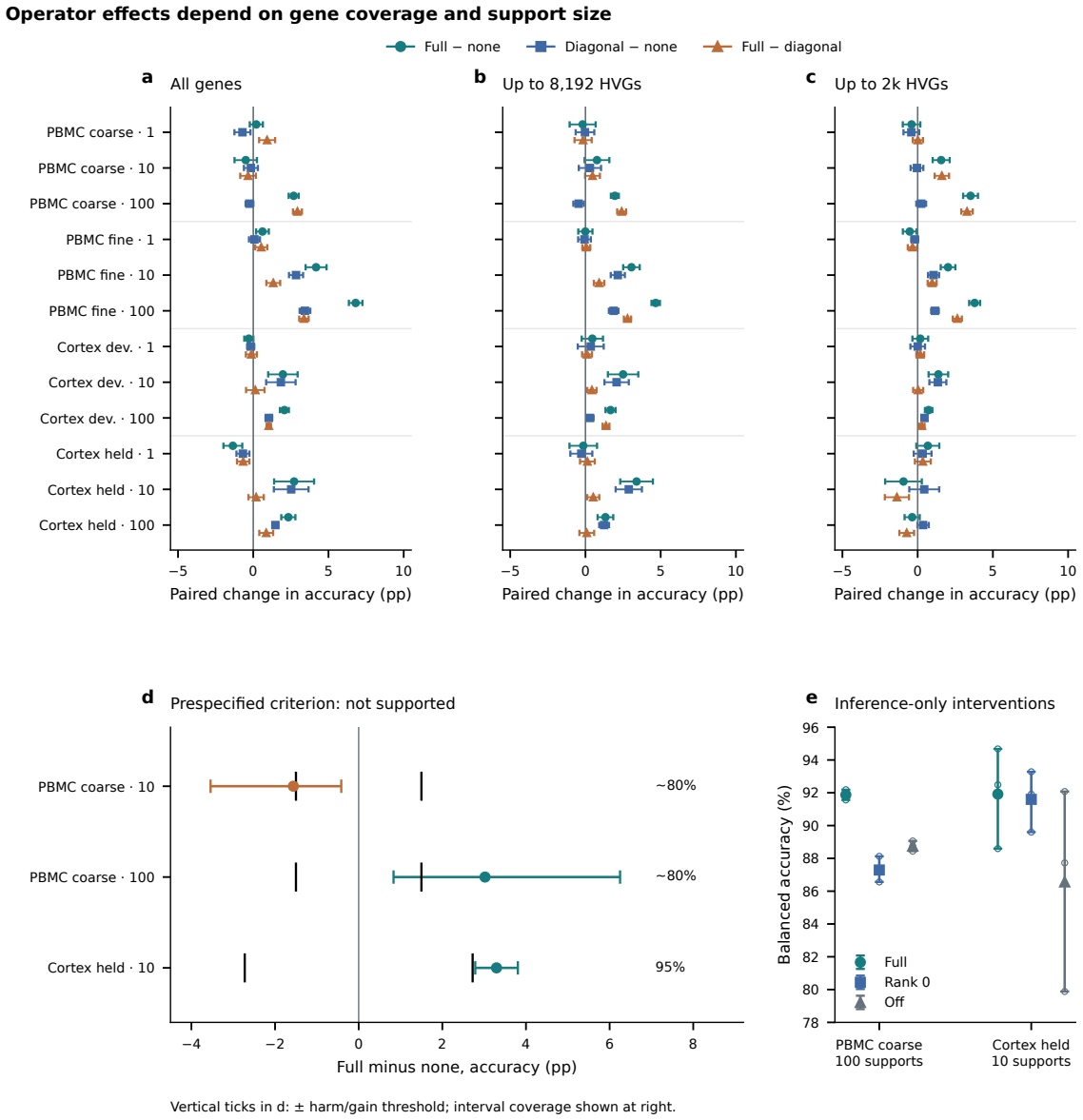


Figure S4: Operator effects depend on support and gene coverage; the prespecified test was not supported. a–c, Separately trained full, diagonal and no-operator balanced-accuracy contrasts using all genes, up to 8,192 HVGs or up to 2k HVGs. Rows specify panel and supports per class. Points and bars show paired means and 95% t intervals over twenty support draws, conditional on one training seed per arm and fixed queries. HVG widths are requested caps. At one support, the operator gate is zero; differences reflect trained-network differences. **d**, Original prespecified test using three frozen draws. Black ticks mark thresholds. Coarse-PBMC ten-support balanced accuracy crossed the −1.50-point harm threshold (−1.5625 points), triggering the overall failed criterion. PBMC bars use four-donor enumeration with approximately 80% coverage; held-study cortex uses a 95% donor bootstrap within study. The later twenty-draw estimates do not overturn that decision. **e**, Full-model rank-zero and operator-off interventions at inference, showing all three draws, means and ranges. These differ from the separately trained arms in a–c. All checkpoints are at update 96,000; no interval measures training-seed variation. pp, percentage points. Source Data retain the exact contrasts, intervals and control observations.

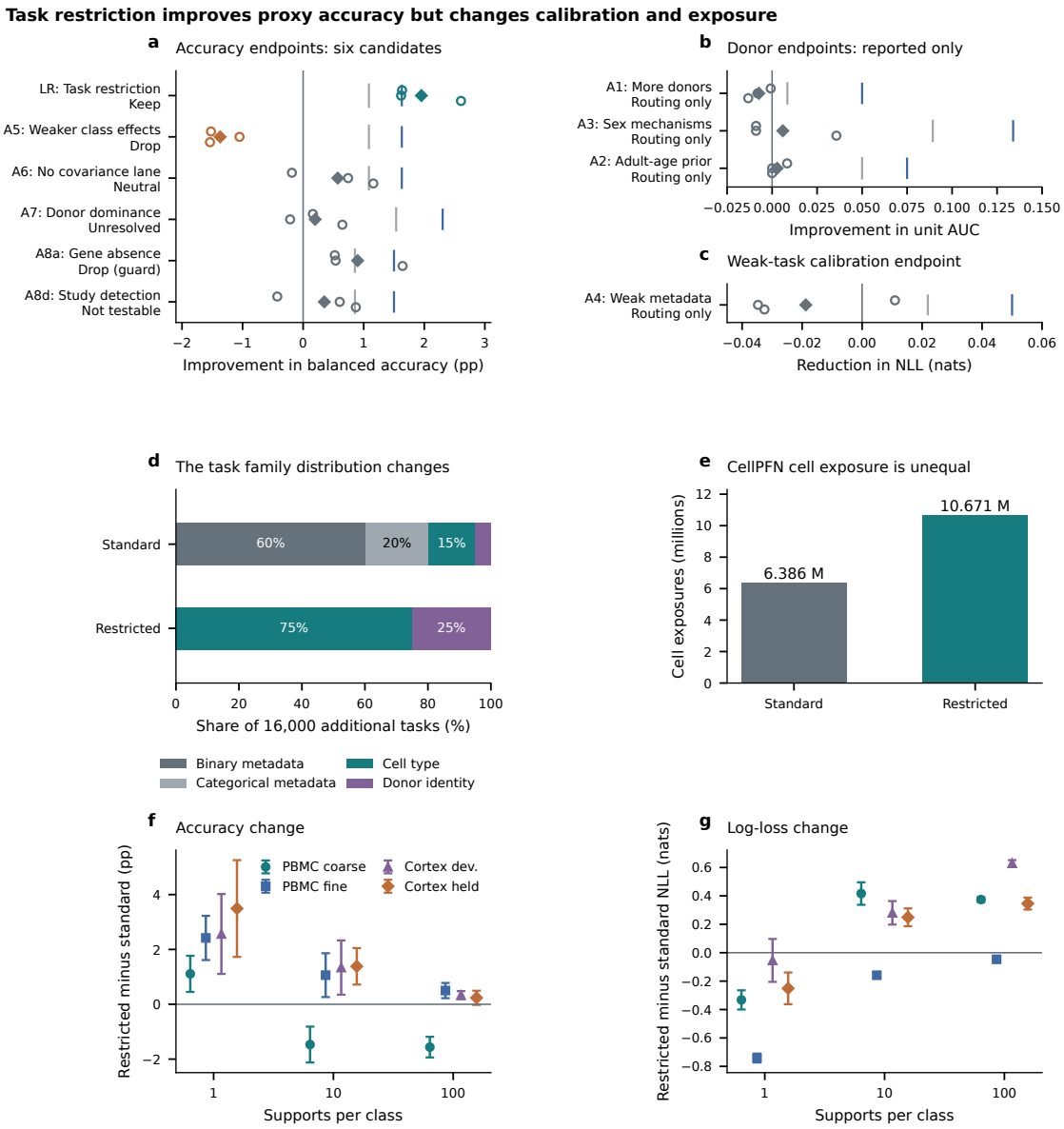


Figure S5: Task restriction improves proxy accuracy with calibration and exposure tradeoffs. **a–c**, All ten prior candidates on the Kumo-small proxy after 10,000 cellular tasks, 625 updates and up to 2k HVGs. Open circles are three paired task-stream replicates from one model seed; diamonds are means. Gray/blue ticks mark the effective noise band/keep threshold. Positive differences favour each candidate against its specified base. Accuracy endpoints differ across candidates; donor AUC and NLL reduction use separate axes. Supplementary Table S5 defines bases and endpoints. Routing-only labels are not efficacy decisions; A8d is not testable because proxy headroom is insufficient. **d,e**, CellPFN continuation over 4,000 updates and 16,000 tasks per arm. Standard training supplied 12,816 metadata, 2,398 cell-type and 786 donor-identity tasks; restriction supplied 12,001 cell-type and 3,999 donor-identity tasks and excluded signed-covariance banks. Cell exposures were 6,386,352 versus 10,671,056 (1.67-fold), including reuse. **f,g**, Restricted minus standard balanced accuracy and NLL at update 100,000 with all genes: paired means and 95% t intervals over twenty support draws. Positive NLL change is worse. Task composition, bank acceptance and cell exposure change together; intervals do not cover training-seed variation. pp, percentage points. Source Data retain full frozen decisions and numerical summaries.

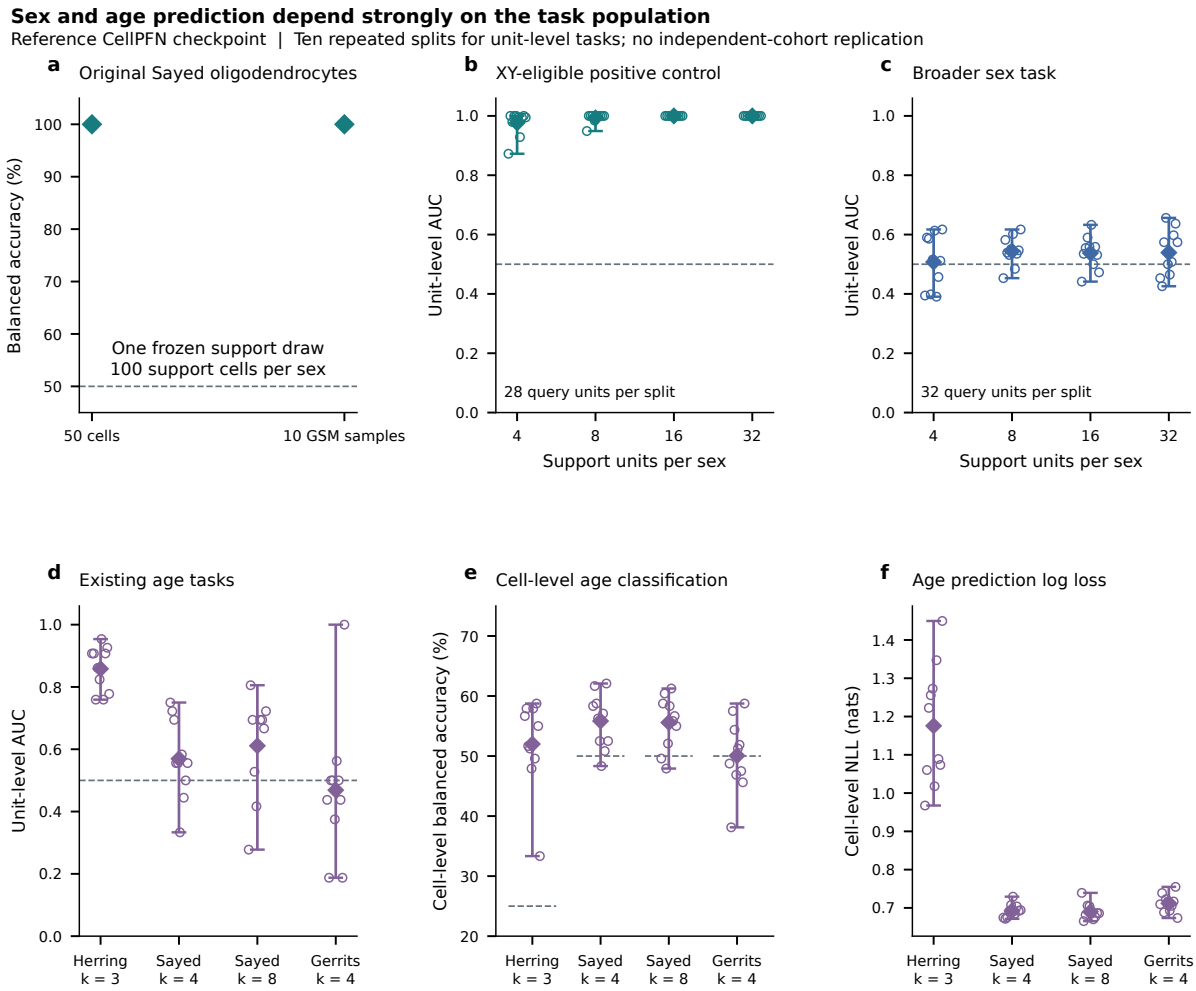


Figure S6: Sex and age performance depends on the task population and target. a, One frozen Sayed oligodendrocyte sex case: 100 support cells per sex from twelve GSM samples and 50 query cells from ten other GSMs. GSM separation does not establish donor separation; no population interval is inferred. **b,c**, Unit-level sex AUC for studies measuring *XIST* and at least four Y-linked genes (positive control; 28 query units) versus the broader task (32 query units). Populations and gene universes differ; eligibility does not identify predictive genes. **d–f**, Age AUC, cell-level accuracy and NLL. Herring excitatory neurons use four age classes (< 1, $1–< 6$, $6–< 16$, ≥ 16 years), three supports/class and twelve query persons. Sayed/Gerrits oligodendrocyte targets split eligible units at the study median, with twelve/eight query units. Herring AUC is macro one-versus-rest; other AUCs are binary. Circles show ten repeated splits; diamonds/bars show means/ranges. Reused units are not independent cohorts. Dashed lines mark chance; NLL depends on class count. All panels use checkpoint 96,000; b–f request up to 8,192 HVGs. Units are resolved persons or Sayed age-by-sex GSM groups; cross-study identity overlap is unresolved. No sex-gene attribution is asserted.

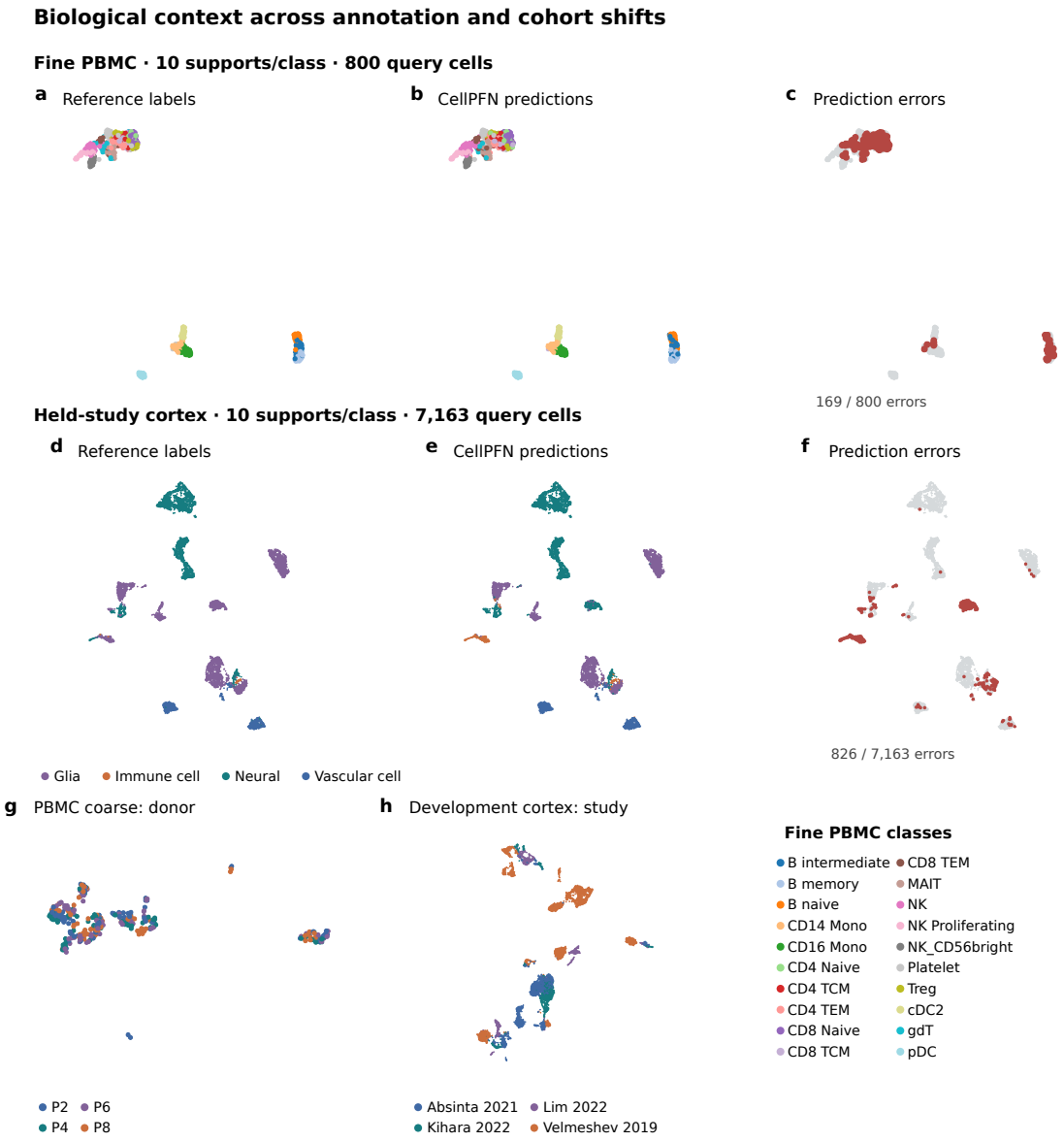


Figure S7: Expanded biological maps. a–c, Fine-PBMC reference labels, CellPFN predictions and errors for all 800 queries at ten supports/class, draw 0. **d–f,** Corresponding views for all 7,163 held-study cortex queries, also at ten supports/class and draw 0. Gray points in error panels are correct predictions; red points are errors. Fine PBMC uses a twenty-category palette; cortex colors match the main biological figure. **g,h,** Coarse-PBMC donor context (320 queries, 100 supports/class) and development-cortex study context (7,769 queries, ten supports/class). These coordinates match the corresponding main-figure and marker maps. No batch integration is applied. All queries are retained. Each case's RNA geometry was fitted once from its exact support/query union without labels: 2k HVGs, standardized expression clipped at ten standard deviations, at most fifty principal components, fifteen Euclidean neighbors and UMAP minimum distance 0.3, seed 42. Display preprocessing does not alter stored classifier inputs or outputs (Methods). Overlays within a case share coordinates; independently fitted cases have no common coordinate frame. These maps show RNA variation and model outputs, not CellPFN embeddings or independent cohort replication. Source data retain identities, coordinates and displayed values. Supplementary Figure S7 continues with panels **i–p**.

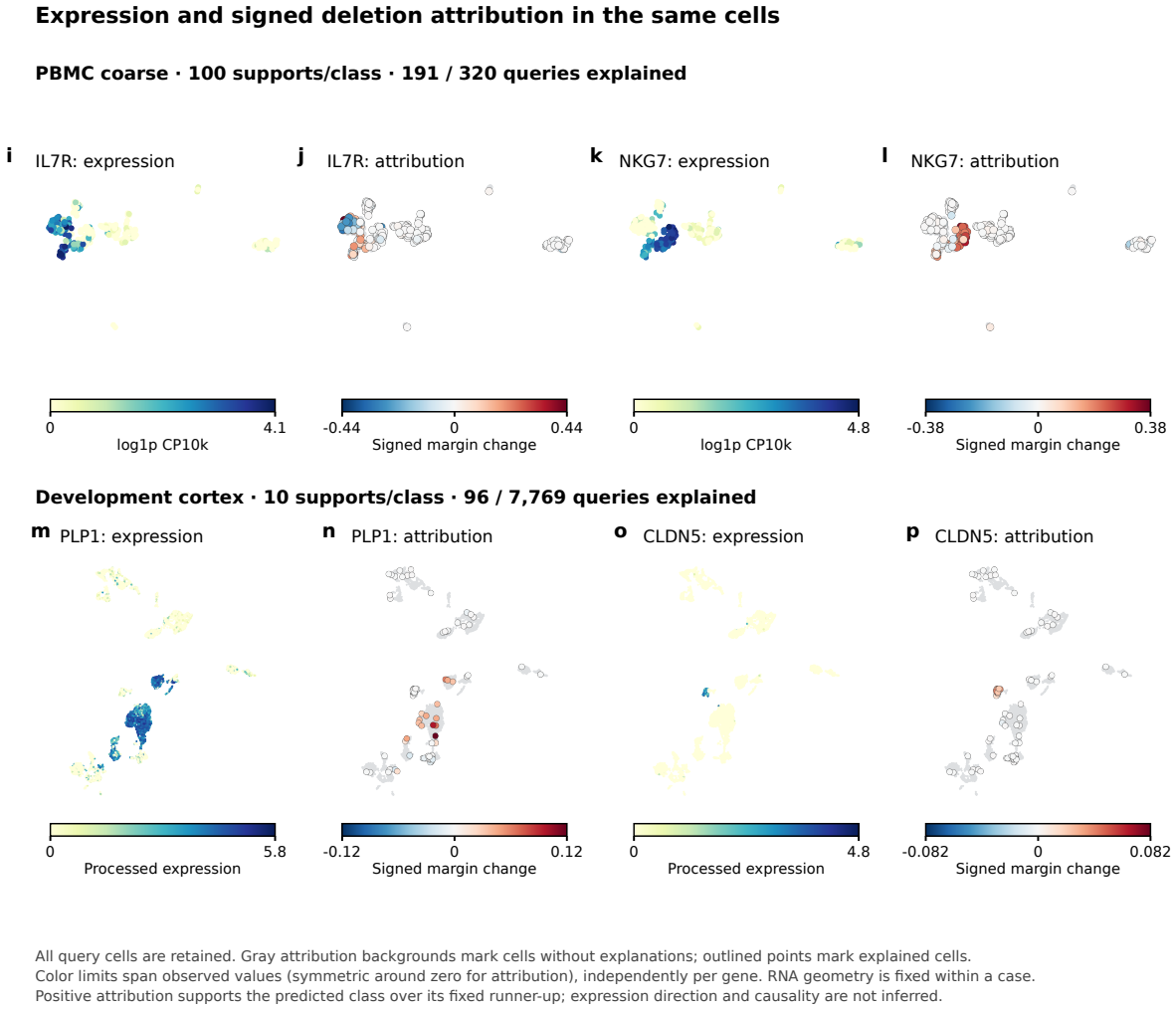


Figure S7 (continued). Expression and signed deletion attribution. i–l, *IL7R*/*NKG7* query expression and corrected DS-LOO attribution in coarse PBMC (320 queries; 100 supports/class, draw 0). **m–p,** *PLP1*/*CLDN5* overlays in development cortex (7,769 queries; ten supports/class, draw 0). Marker choices preceded map generation. PBMC expression is log1p counts per 10,000; cortex retains supplied processed values. Coordinates match panels **g,h** and the main biological figure. Expression uses each gene's full observed range; attribution limits are symmetric around zero, independently per gene. Positive attribution means that the observed gene favors the predicted class over its fixed runner-up; negative values oppose that margin. Neither sign establishes regulation or causality. Pale-gray attribution backgrounds mark unexplained queries; outlined colored points show all explained cells, including exact-zero scores, enlarged solely for visibility. Explanations cover 191/320 PBMC and 96/7,769 cortex queries, selected among correctly classified cells, and therefore do not characterize every query. Every query remains visible. RNA geometry and preprocessing are described in panels **a–h** and Methods; the renderer performs no additional normalization, embedding or model inference. Source data preserve cell identities, coordinates and displayed expression/attribution values.

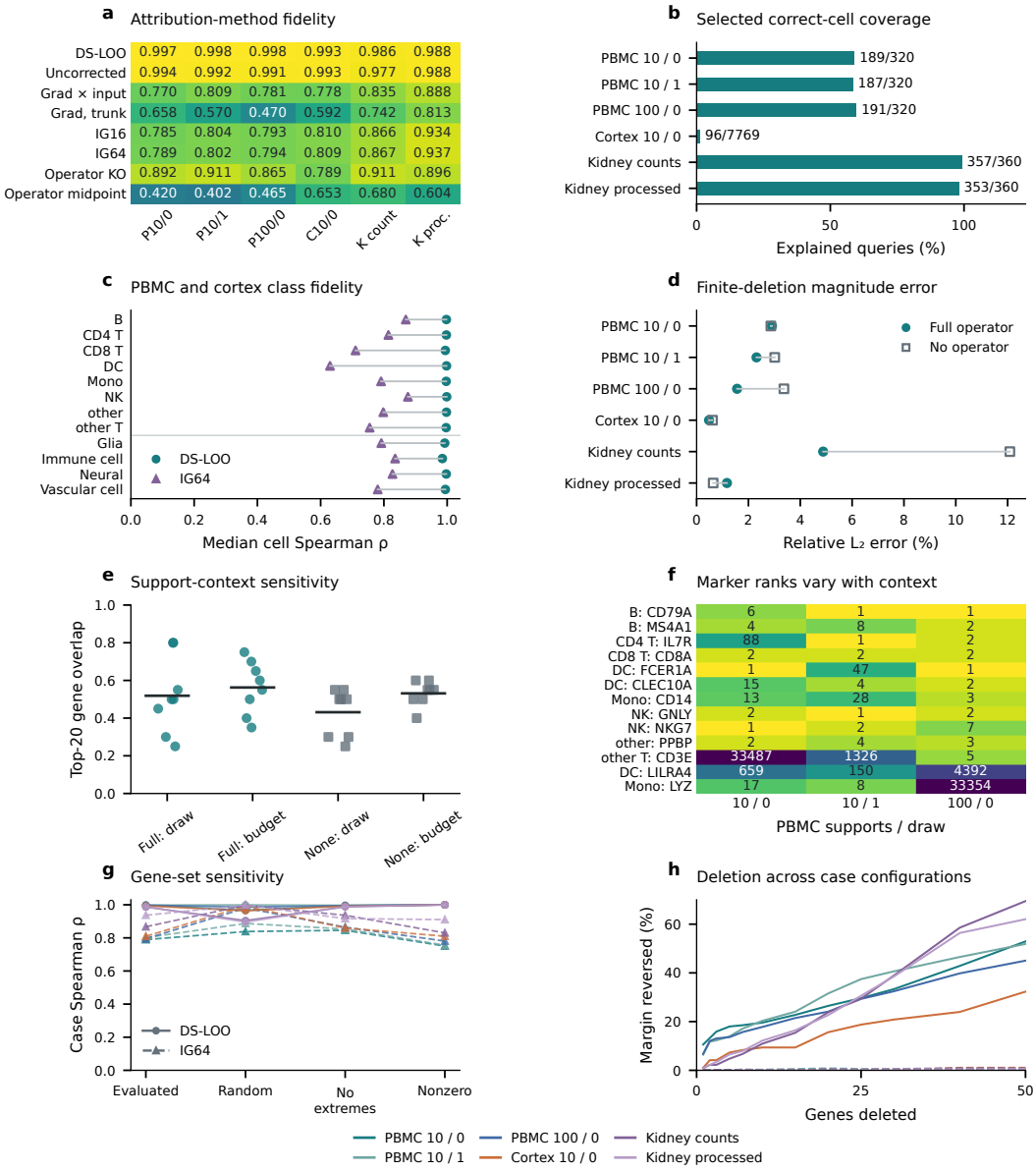


Figure S8: Explanation fidelity, coverage and support-context sensitivity. a, Within-cell Spearman correlation with exact query-only gene deletion, summarized within and across classes, for eight attribution variants; independent recomputation on the original evaluated set. Colors span zero to one. **b,** Explained/all query cells, selected for correctness in stored and float32 predictions (Methods). **c,** Per-class median within-cell fidelity for PBMC at 100 supports and development cortex at ten supports; divider separates datasets. **d,** Corrected DS-LOO relative L_2 error, computed per class and summarized by the median. Full and no-operator models use their own selected cells. **e,** Top-20 overlap after changing support draw or budget. Points are eight classes; horizontal marks are means. Budget changes are not seed-only comparisons. **f,** Every PBMC marker check in Supplementary Table S8, including nonhits. Lower ranks indicate higher importance; color is logarithmic. **g,** Fidelity across validation subsets; "Nonzero" means absolute exact effect $> 10^{-4}$ per cell. Solid circles, DS-LOO; dashed triangles, IG64. **h,** Original runner-up margin reversal after ranked deletion (solid) or pooled random-expressed deletion (dashed). Colors in **g,h** identify cases. Labels give supports/draw; kidney modes share cells. Values are descriptive, without confidence intervals. One-shot and held-study-cortex validation are unavailable. Zero deletion effects do not assess absence information; fidelity does not establish biological causality.

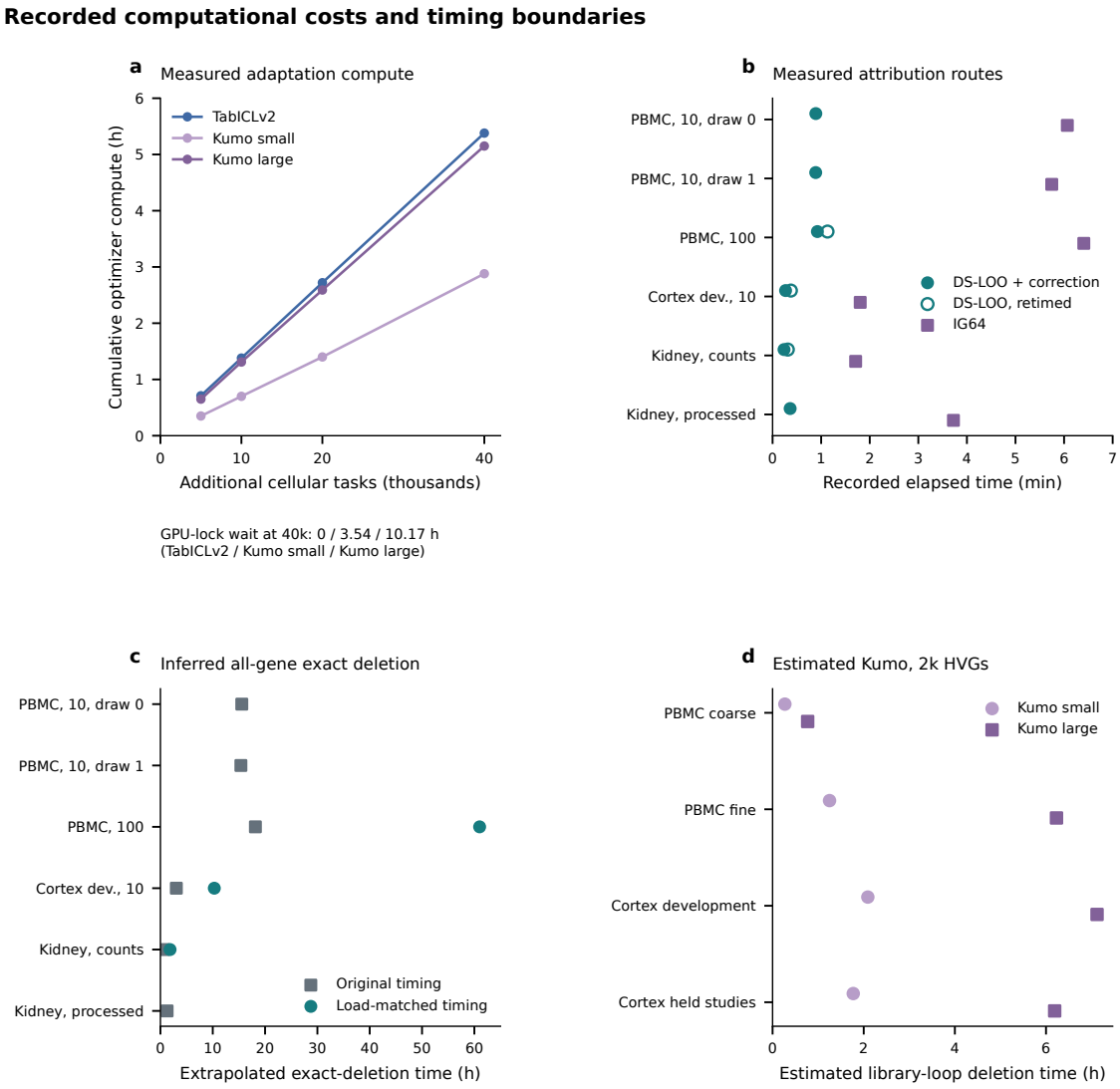


Figure S9: Recorded computational costs and measurement boundaries. a, Cumulative measured optimizer compute for continued training on one NVIDIA H100 80-GB GPU. Cellular task counts exclude 25% additional generic replay. GPU-lock waiting is separate. Costs exclude released-model pretraining, startup, hyperparameter screens, evaluation, checkpoints and real-data scoring. Each arm has one model seed. **b,** Measured corrected DS-LOO and 64-step integrated-gradients (IG64) times for six CellPFN cases. Filled points use original timings; open circles show available load-matched DS-LOO reruns. Cases use selected explained cells and their exported attribution scope, not all queries. Attribution uses fp32. GPU sharing varied, and IG64 was not retimed under every DS-LOO condition. No controlled speedup or uncertainty interval is inferred. **c,** Inferred exact-deletion times over all genes of those explained cells, extrapolated from measured single-gene work. Squares/circles distinguish original/load-matched contexts. Full all-gene runs were not measured. **d,** Estimated Kumo deletion time for 2k HVGs under the library-as-is loop: recorded median forward time multiplied by required calls. These fp16 estimates are not completed deletion runs. Kumo and CellPFN differ in query counts, genes, scope, precision and timing conditions; panels **b–d** do not establish controlled model-speed comparisons. No numerical simulator-throughput claim is made.